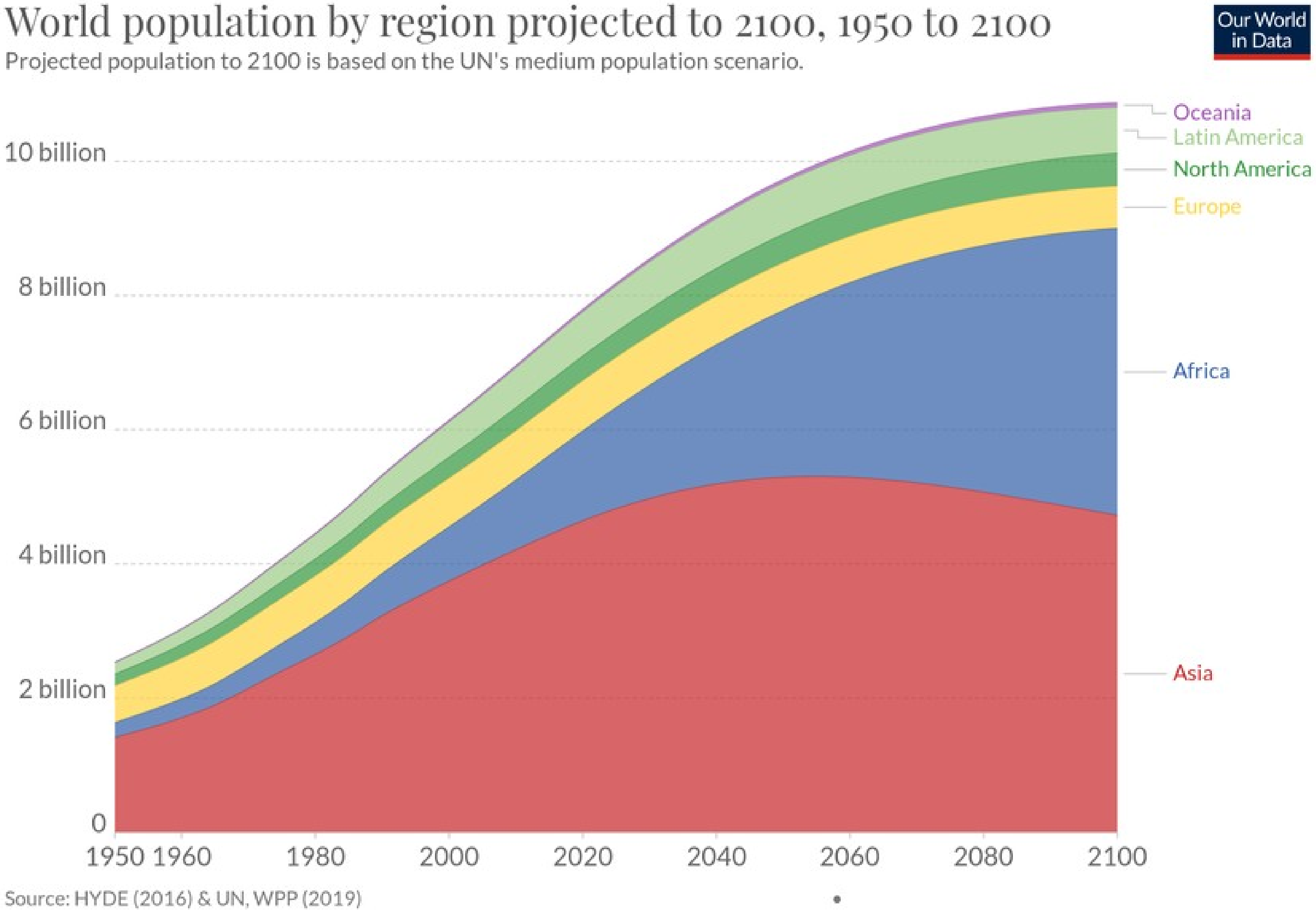
PUBP 233
INTRODUCTION TO DEMOGRAPHY (POPULATION AND SOCIETY)

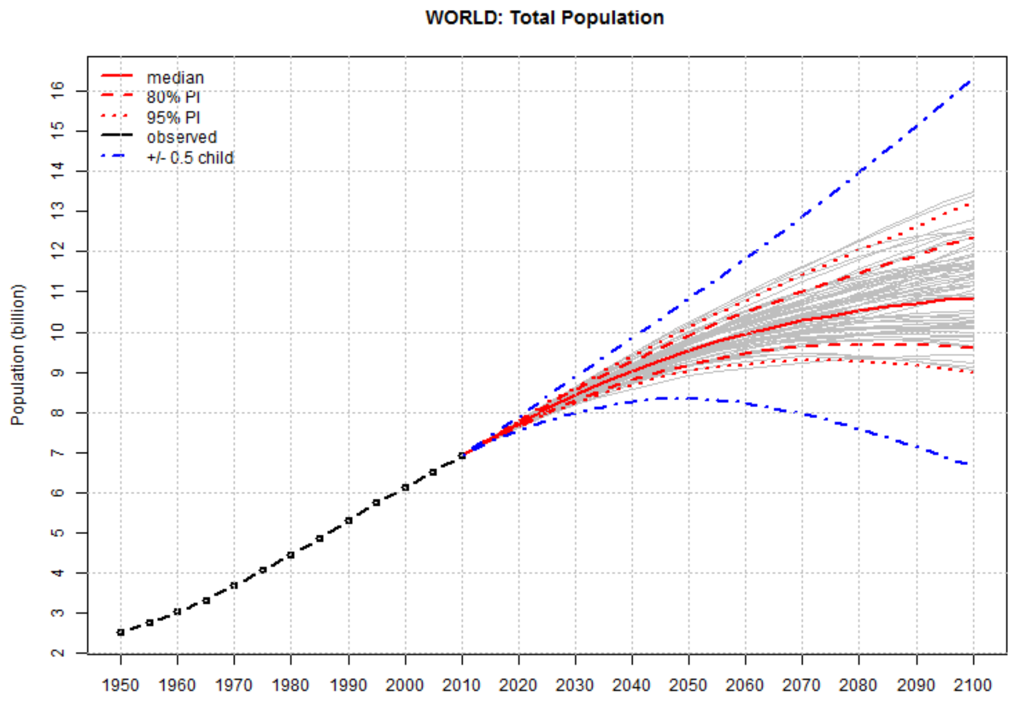
Shivakumar Jolad

Week 14: Population Projection

Population Projection



PROJECTIONS



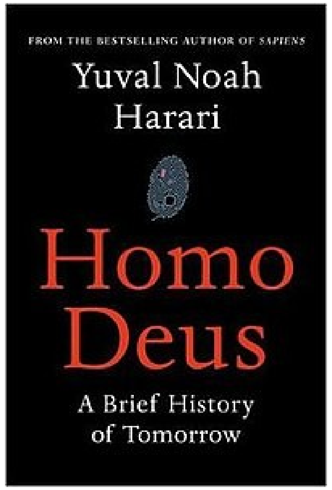
Analyze the **future trajectory of population change**

Sought after by all: Government and Private sector
- Examples?

Demographic forecasts are fundamental to any form of social, economic or business planning.

Can help to **defy myths about population growth, separate myths from reality**

Can help in assessing - political and social consequences of population change in future

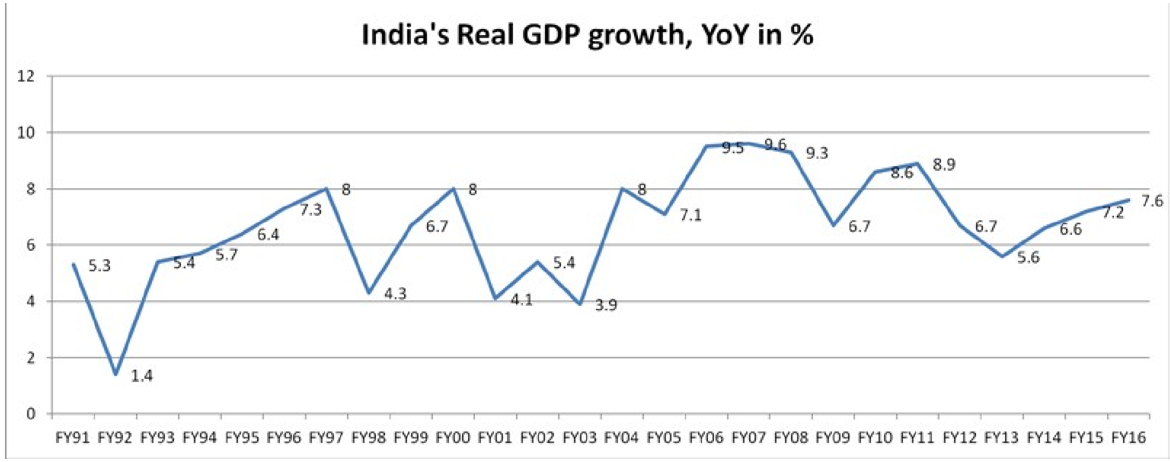


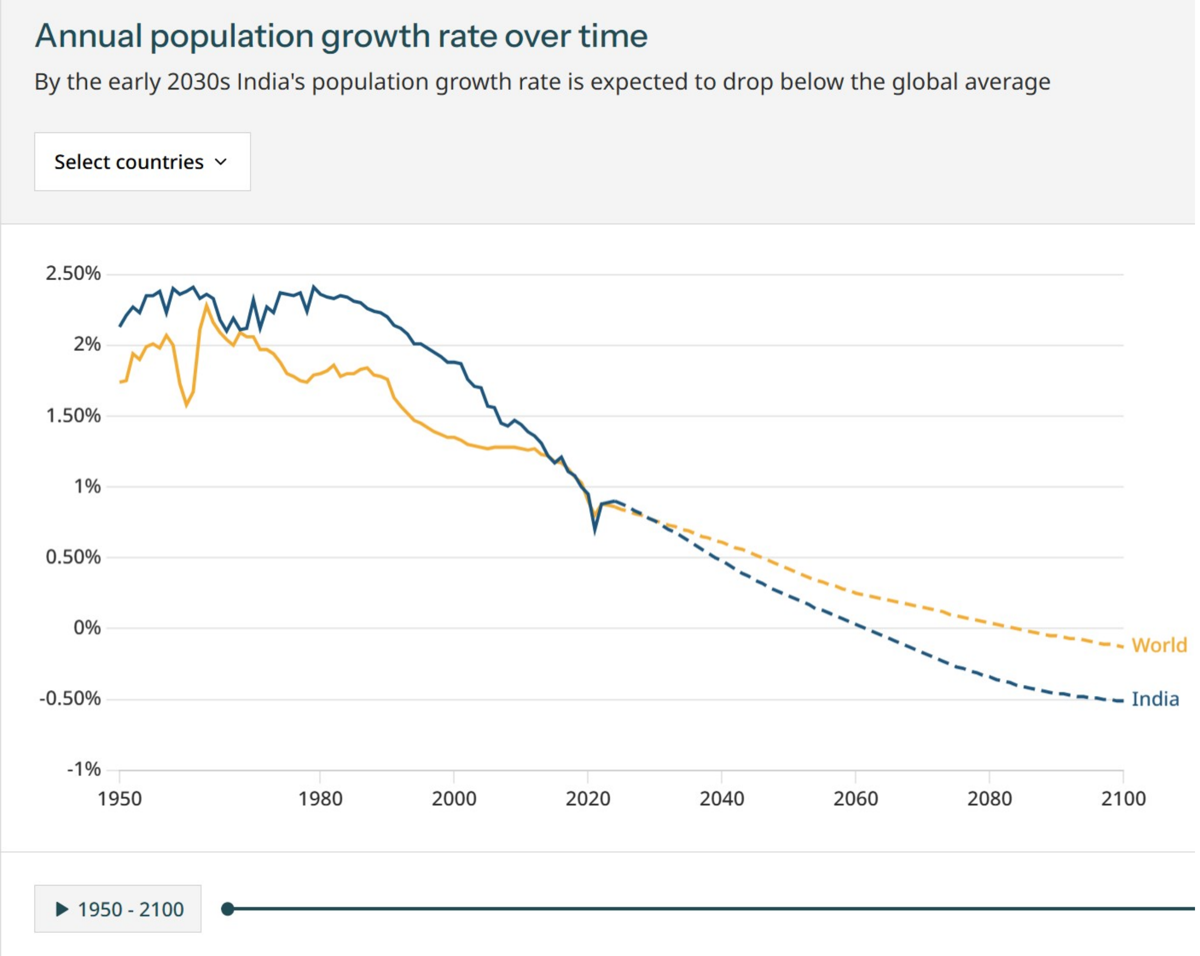
PREDICTING FUTURE?

Human populations have two fundamental characteristics that reduce uncertainty about how they will develop in the future:

- A **substantial overlap exists between the current population and the future population.**
- Fertility and mortality patterns do not rapidly change abruptly - Reasonable **understanding of fertility and mortality trends** already exists
- **Every year that passes we all get exactly one year older until we eventually die.**

ECONOMIC GROWTH RATE AND POPULATION GROWTH RATE

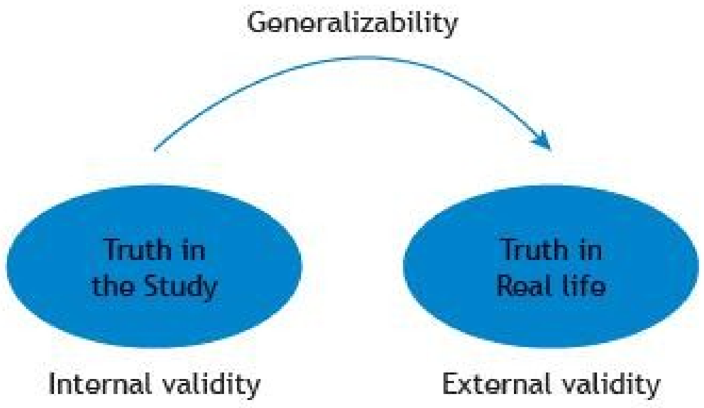




PROJECTIONS AND FORECASTS

Population projection: A computational procedure to *calculate population size and structure at one time from population size and structure at another,* together with a specification of how change takes place during the interim period.

- Drivers: Fertility, Mortality and migration
- **Consistency among demographic variables;** should obey demographic accountancy !
- Subject to **internal validity**



Population forecast: A forecast of future population can be defined as a projection based on assumptions that are predictive and considered to yield the most probable estimates of the future development of a population.

- Subject to **external validity**
- All forecasts of the population are projections, not all projections are forecasts.

METHODS FOR POPULATION PROJECTIONS

Total methods: Calculate trends in the size of the population as a whole using a mathematical model of population growth.

- Distribute the total into sub-groups in ratio to the current structure of the population or an extrapolated forecast of its structure.
- Also known as **ratio** methods of projection.
- Ex models: Zero population growth. Arithmetic growth, exponential growth, logistic growth

Cohort component methods: project each age group, sex and other category of interest separately. Then aggregate the results to obtain the total population.

- Cohort emphasizes that an age group is made up of people born at the same time who go through life together.
- The size of a cohort at one age (and date) is strongly predictive of its size at other ages (and dates).

TOTAL METHOD

Approach

- Select an appropriate model of the growth process
- estimate the parameters of the model from past estimates of the population
- extrapolate the fitted curves and read off the projected population

Examples:

- Zero population growth
- Arithmetic growth
- exponential growth
- logistic growth

WHAT IF GROWTH REMAINED CONSTANT?

If the population growth remained at 2% annually :

- Less than 700 years, there would be one person for every square foot
- In less than 1200 years, human population would outweigh the earth !
- In less than 6000 years, the mass of Humanity would form a sphere expanding at the speed of light

The Power of Geometric Growth

$$\frac{dN}{dt} = rN$$

$$N(t) = N_0 r^t$$

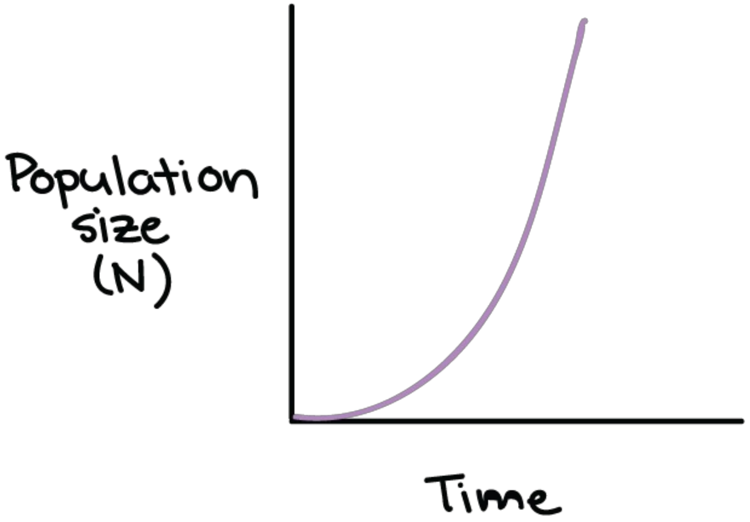
Coale, A. J. (1974). *The History of the Human Population. Scientific American, 231(3),*

$$\frac{dN}{dt} = rN$$

Exponential growth

Per capita growth rate (r) doesn't change, even if pop. gets very large.

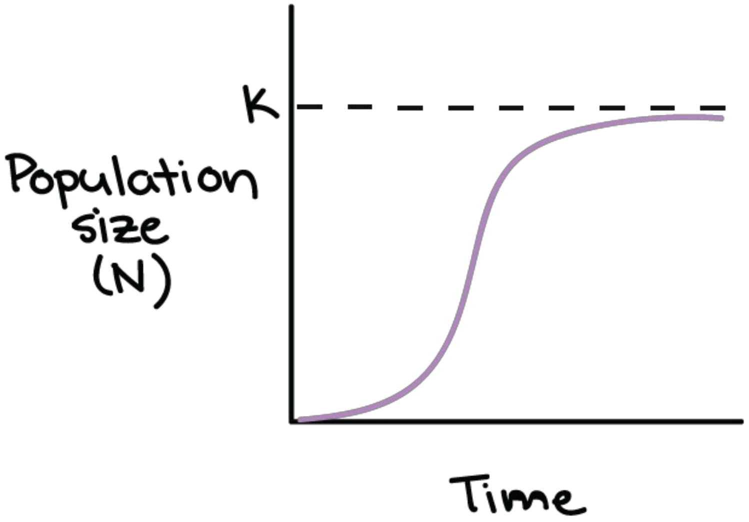
$$\frac{dN}{dt} = r_{max} N$$



Logistic growth

Per capita growth rate (r) gets smaller as pop. approaches its max. size.

$$\frac{dN}{dt} = r_{max}\left(\frac{K-N}{K}\right) N$$

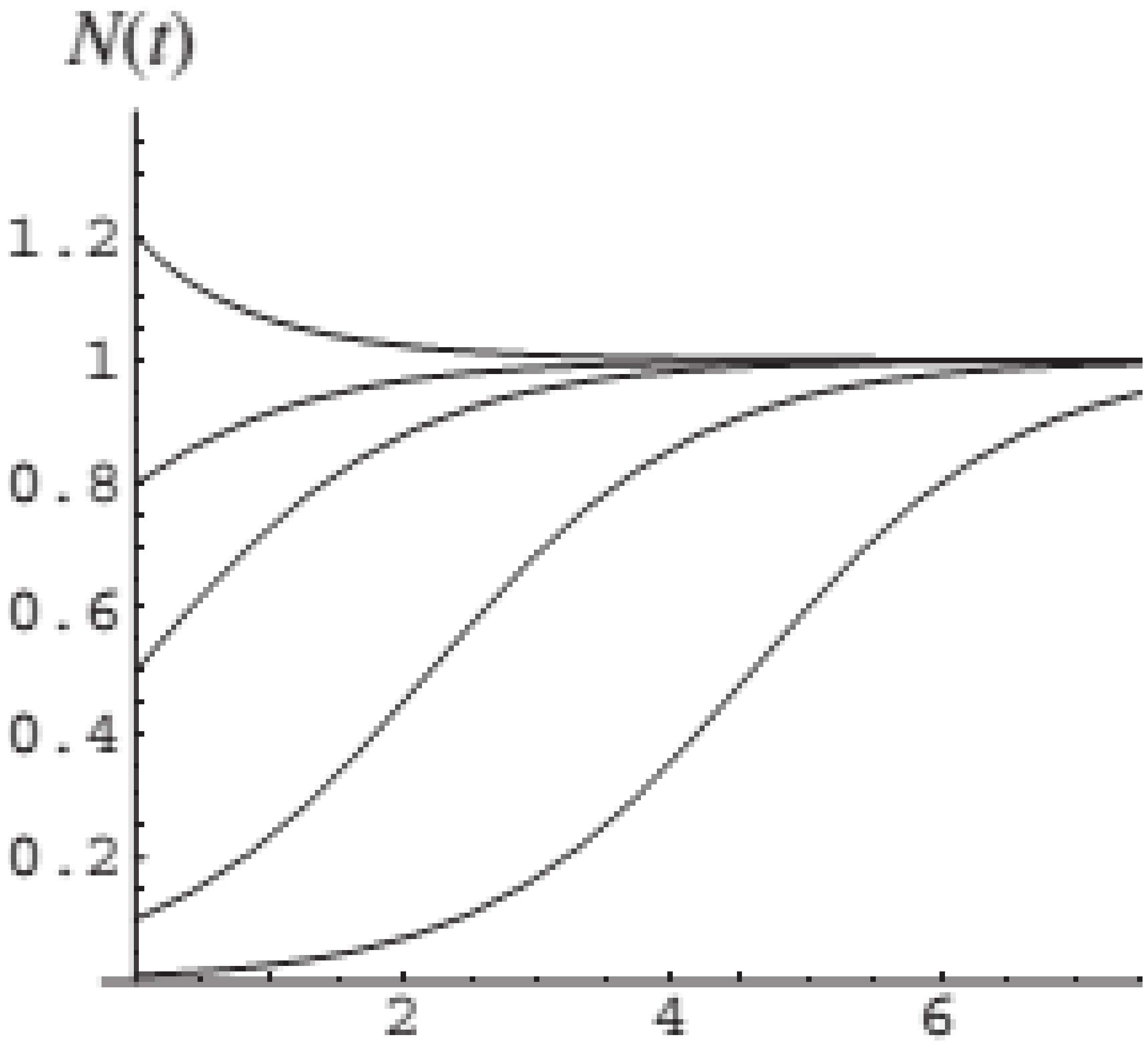


https://www.khanacademy.org/science/ap-biology/ecology-ap/population-ecology-ap/a/exponential-

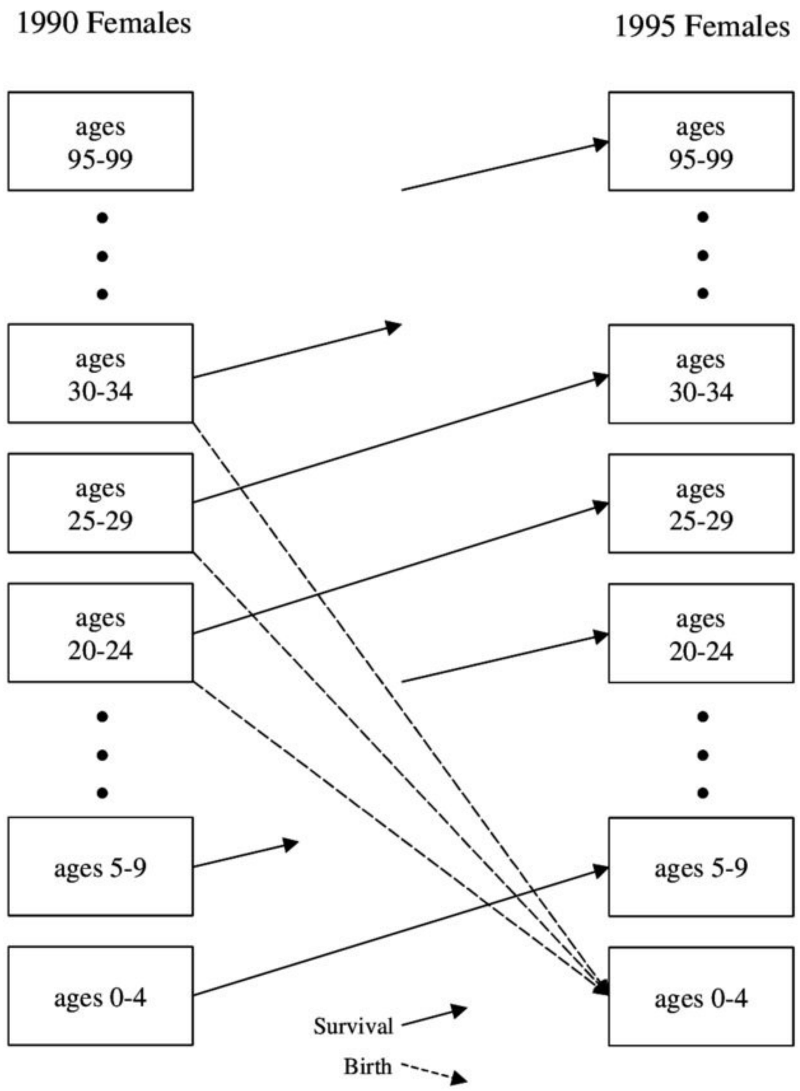
TOTALS METHODS- LOGISTIC GROWTH

$$\frac{dN}{dt} = rN\left(1 - \frac{N}{K}\right)$$

$$N(t) = \frac{K}{1 + e^{-r(t-t_o)}}$$



http://gi.ics.nara-wu.ac.jp/~takasu/lecture/graduate_course/L9_logistic-growth.pdf



COHORT-COMPONENT POPULATION PROJECTIONS

Model the age-sex structure of populations and not just their size

Model the components of demographic change - **fertility, mortality, and migration** – and not just population growth.

Basic ideas

- **Population balance equation**

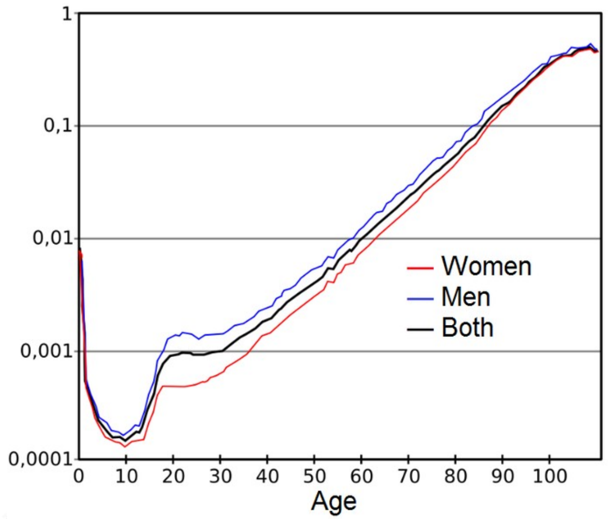
$$N(t+n) = N(t) + B(t,n) - D(t,n) + IM(t.n) - OM(t,n)$$

- **Age**: Every year people age by one year!
- **Stable Rates:** Fertility, Mortality, Migration

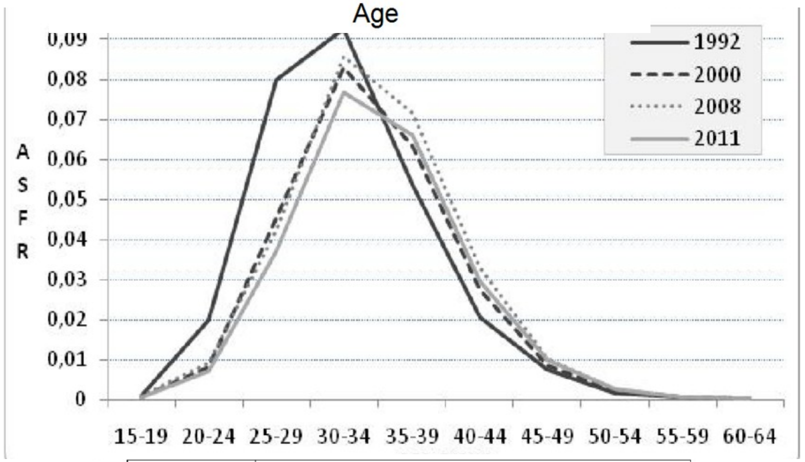
A Guide to Global Population Projections, February 2001, Demographic Research 4(8):203-288

DISCRETE MODELS- COHORT COMPONENT

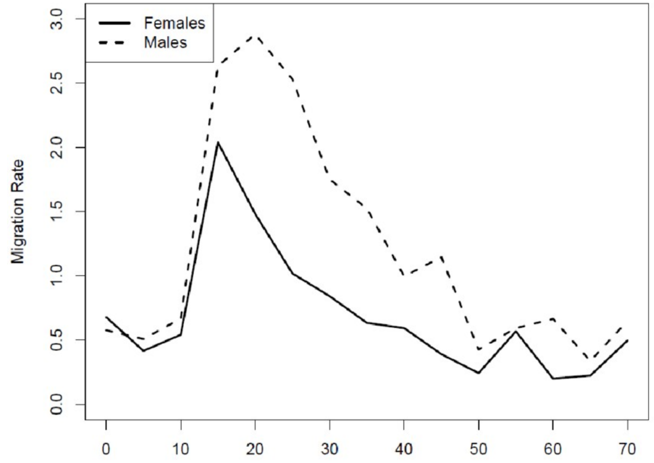
- **Separate projection** for each sex; Focus on Female dominant models
- **Age group – same interval,** except for last interval
  - 0-1 years and 1-4 years clubbed to form 0-4 years
- From **Life table:** Survivorship curve $l(x)$ and person years $L(x, n)$
- **Fertility rates:** 15-49 years. Age intervals in life table and fertility table should be same.
- Main steps (for single sex):
  - **Project every age coho**rt for one projection interval at a time
  - Calculate the births during this interval and add in the newly-born children
  - Adjust for **migration**



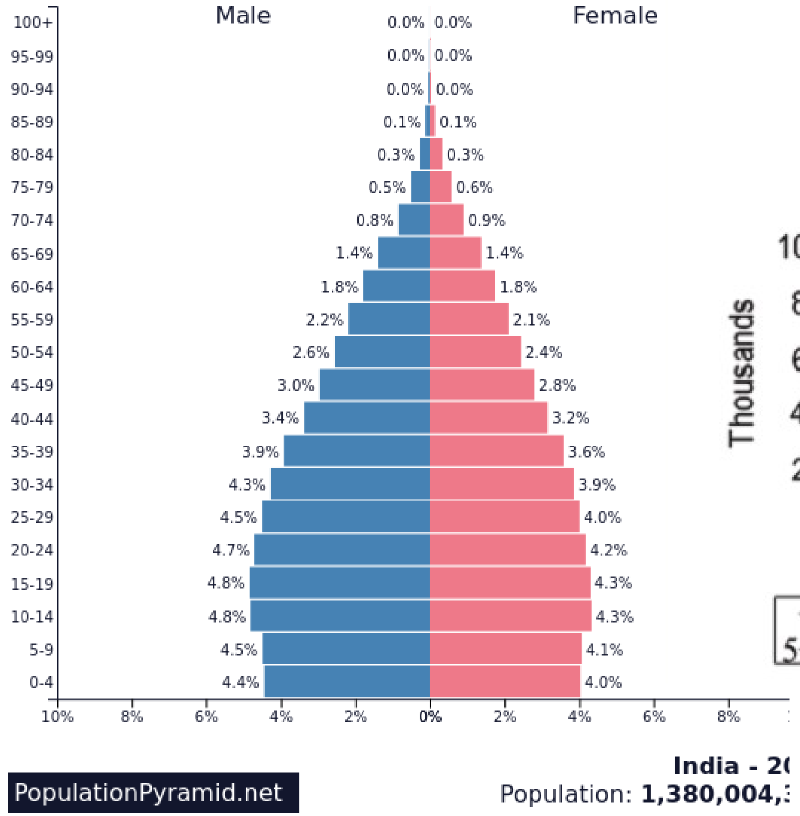
Women
Men
Both
Age



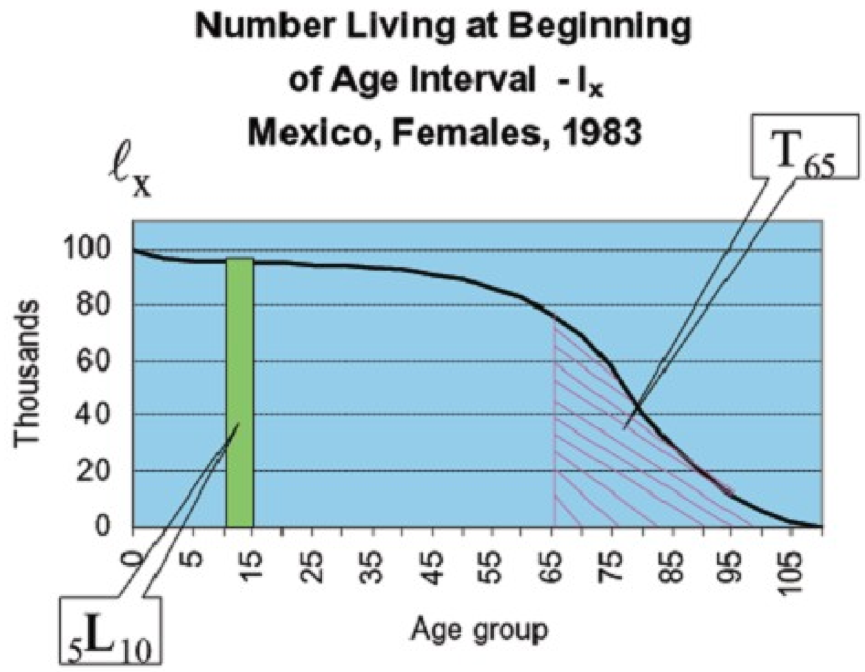
1992
2000
2008
2011
ASFR
15-19 20-24 25-29 30-34 35-39 40-44 45-49 50-54 55-59 60-64



Females
Males
Migration Rate



Male
Female
PopulationPyramid.net
India - 20
Population: 1,380,004,3



Number Living at Beginning
of Age Interval - l_x
Mexico, Females, 1983
ℓ_x
T_{65}
Thousands
$_5L_{10}$
Age group

STEPS

Project	project every age cohort for one projection interval at a time
Calculate	calculate the births during this interval and add in the newly-born children
Adjust	adjust for migration
Repeat	repeat the procedure to project the population to the end of the next interval

http://papp.iussp.org/sessions/papp101_s10/PAPP101_s10_010_010.html

DATA REQUIRED FOR CC PROJECTIONS

Base year population subdivided by age and sex

Sex-specific life tables for each projection interval in the projection period (mortality)

Age-specific fertility rates for each projection interval in the projection period

Age- and sex-specific net migration for each interval in the projection period (unless one is assuming that the population is closed to migration).

http://papp.iussp.org/sessions/papp101_s10/PAPP101_s10_010_010.html

CALCULATIONS REQUIRED IN A POPULATION PROJECTION

Calculate **how many births will occur during current projection interval and divide them into boys and girls.**

Calculate **how many of these births of each sex will remain alive at the end of the projection interval and adjust for net migration** into the youngest age group.

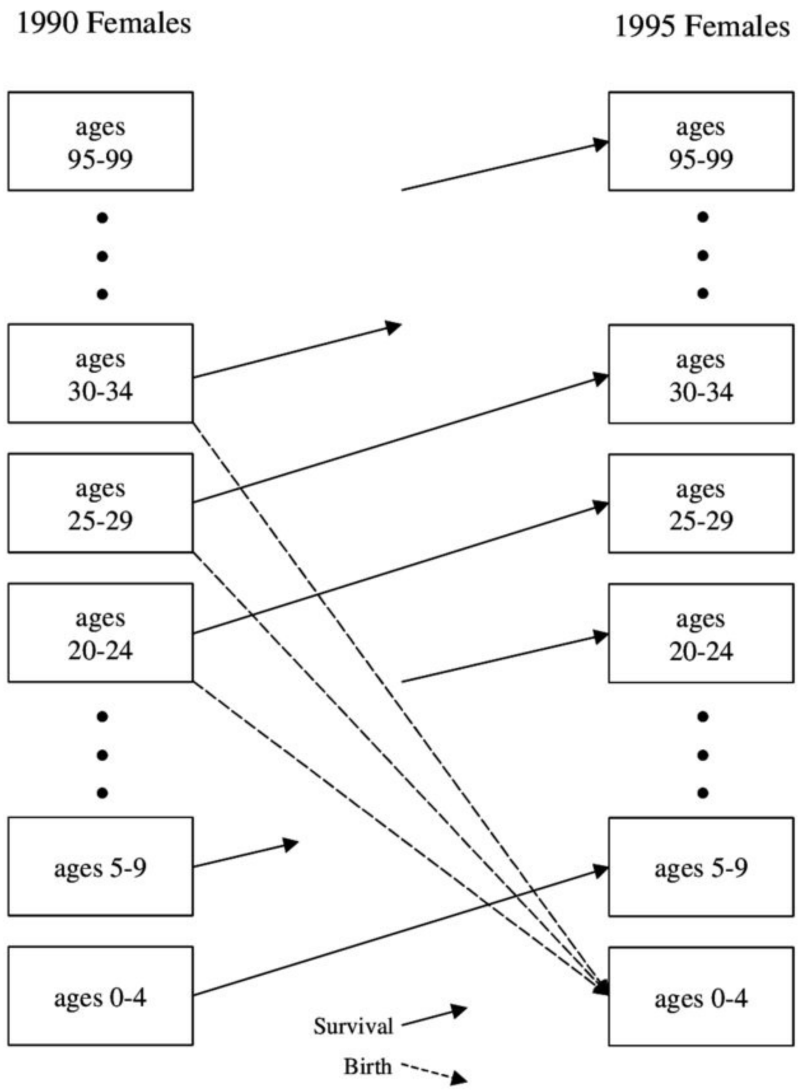
Either add the immigrants to each age cohort and subtract the emigrants or, more simply, **add net migrants to each age cohort**

Repeat the calculations for the next projection interval.

http://papp.iussp.org/sessions/papp101_s10/PAPP101_s10_010_010.html

EXAMPLE: SRI LANKA-2010

Age group	Female population in 2010 ('000s)	Life table person-years 5Lx (e0=77)	Net migrants ('000s)	Age-specific fertility rates	Births by age of mother
0–4	929	486743	-2.4		
5–9	856	484440	-2.6		
10–14	764	483819	-2.3		
15–19	797	483213	-2.5	0.0222	86.4
20–24	818	482354	-4.1	0.0938	377.4
25–29	873	481210	-5.1	0.1502	632.3
30–34	821	479708	-3.5	0.1155	487.3
35–39	733	477656	-2.1	0.0533	206.3
40–44	732	474667	-1.3	0.0138	50.3
45–49	689	470169	-0.8	0.0012	4.2
50–54	641	463275	-0.8		
55–59	527	452580	-0.8		
60–64	454	435860	-0.7		
65–69	338	409120	-0.5		
70–74	252	367295	-0.3		
75–79	170	307729	-0.2		
80–84	102	228984	0		
85+	66	231178	0		
Total	10562		−30.0	2.25	1844.2
Female births					**899.6**



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CCM Model implementation – Example

A simplified Cohort-component method projection of the female population of "Westlands", from 2010 to 2015

Still need to estimate youngest cohorts based on fertility data

Population values are expressed in thousands

1. Survive base population by age and sex

First age group (0-4 → 5-9)

= **1,084** * (459092 / 469097) = 1,061

$$S_{x,x+n} = \frac{{}_nL_{x+n}}{{}_nL_x}$$

2. Adjust for migration

= 1061-0.9 = **1,060**

3. Survive the open-ended age group:

$${}_\infty P_{85}(t+n) = [{}_5P_{80}(t) + {}_\infty P_{85}(t)] \times \frac{T_{85}}{T_{80}}$$

= **(18+6)** * 39221 / (39221 + 91395) = **7**

Age group	Population in 2010	Person-years lived - ${}_5L_x$ (e_0=64)	Net migrant population	Population in 2015
0-4	1,084	469097	-2.1	
5-9	922	459092	-0.9	1,060
10-14	793	454840	-0.8	912
15-19	706	450962	-4.0	782
20-24	623	445347	-6.9	690
25-29	528	438622	-6.4	607
30-34	442	431379	-4.7	515
35-39	355	423529	-3.1	431
40-44	277	414509	-2.0	345
45-49	223	403997	-1.3	268
50-54	181	390632	-0.8	215
55-59	145	372490	-0.5	172
60-64	111	346186	-0.3	134
65-69	92	306649	-0.2	98
70-74	67	248840	-0.1	74
75-79	41	172444	-0.1	46
80-84	18	91395	0.0	22
85+	6	39221	0.0	7
Total	6,613		-34.4	

BIRTHS

Calculate the **average number of women in each fertile age group** from start to end of projection interval

Multiply the result **by th**e **age-specific fertility rate for that age group** and then by the **number of years in the projection intervals** to estimate the number of births to women in that age group over the entire interval t to $t+n$

Sum these counts of births over all the fertile age groups to get the total number of births occurring between t **and** $t+n$**.**

BIRTHS CALCULATION

Average Women population (x,5) between years t and t+5

$$B(t) = \sum_{x=15}^{45} F(x,5) \boxed{\left(\frac{N^t(x,5) + N^{t+5}(x,5)}{2}\right)} 5$$

n=5; Five years

Births by Women in 15-49 year age group

Female Children born

$$B^F(t) = B(t) \times \frac{1}{1+SRB}$$

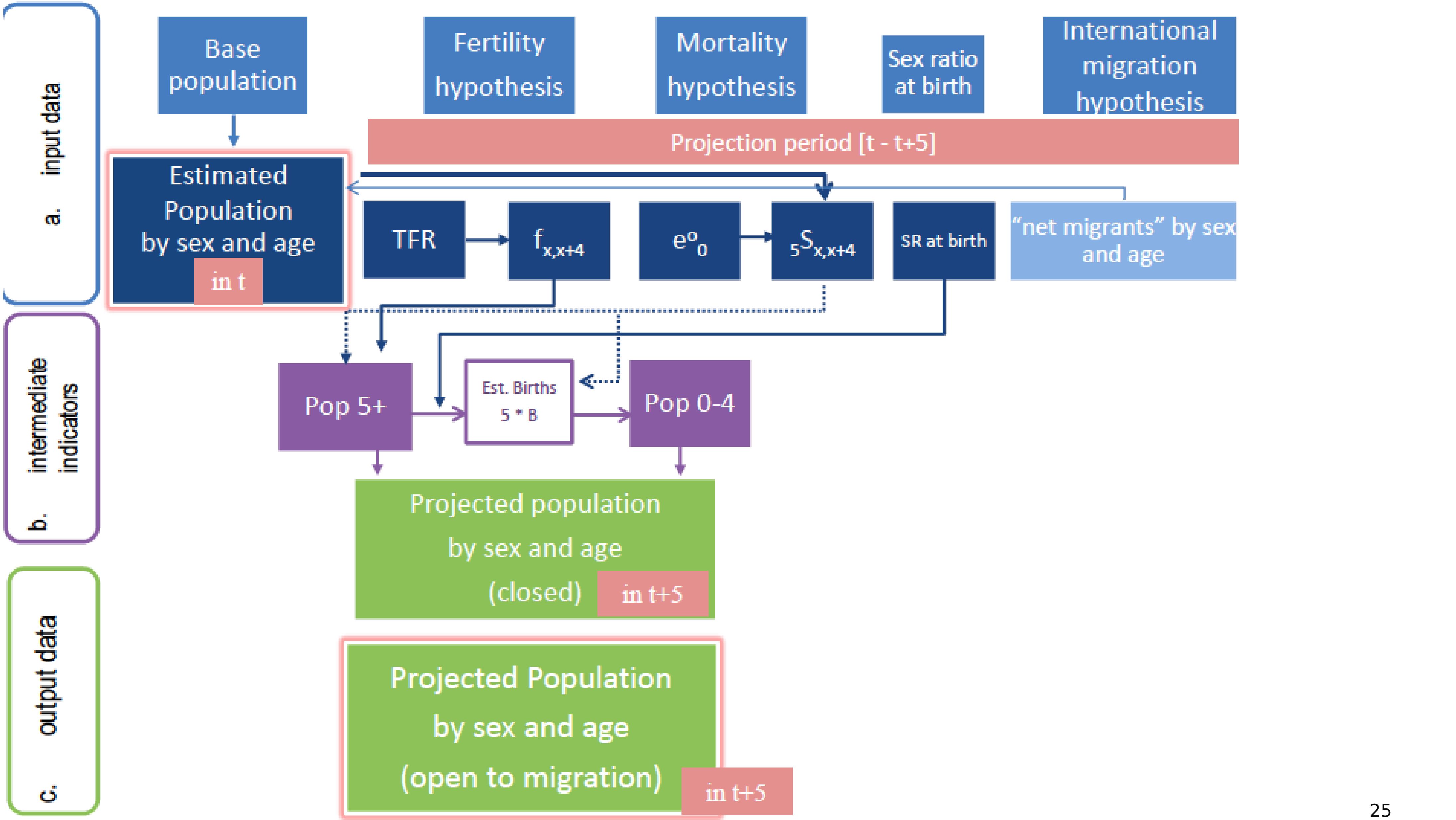
Male children born

$$B^M(t) = B(t) \times \frac{SRB}{1+SRB}$$

SRB– sex ratio at Birth --> 1.05

$$S(0,5) = B^F(0) \times \frac{L(0,5)}{5\ l(0)}$$

Every year l(0) children are born in Life table !


a. input data
b. intermediate indicators
c. output data
Base population
Fertility hypothesis
Mortality hypothesis
Sex ratio at birth
International migration hypothesis
Projection period [t - t+5]
Estimated Population by sex and age
in t
TFR
$f_{x,x+4}$
e^o_0
$_5S_{x,x+4}$
SR at birth
"net migrants" by sex and age
Pop 5+
Est. Births
5 * B
Pop 0-4
Projected population by sex and age (closed)
in t+5
Projected Population by sex and age (open to migration)
in t+5

CCM Model implementation – Example

A simplified Cohort-component method projection of the female population of "Westlands", from 2010 to 2015

4. Calculate the total number of births during interval

- By multiplying projected age-specific fertility rates (ASFRs), with the average number of women in each reproductive age group → an estimation of the surviving women who gave birth during interval

$B_{15\text{-}19} = (706+782)/2 * 0.0967 =$ 360

- Sum up births by age of mother to obtain total births

$B = 2{,}728$

$$B^F(t) = B(t) \times \frac{1}{1+SRB}$$

Age group	Population in 2010	Population in 2015	Age-specific fertility rates	Births by age of mother
0-4	1,084			
5-9	922	1,060		
10-14	793	912		
15-19	706	782	0.0967	360
20-24	623	690	0.2089	686
25-29	528	607	0.2405	682
30-34	442	515	0.2239	536
35-39	355	431	0.1626	320
40-44	277	345	0.0780	121
45-49	223	268	0.0192	24
50-54	181	215		
55-59	145	172		
60-64	111	134		
65-69	92	98		
70-74	67	74		
75-79	41	46		
80-84	18	22		
85+	6	7		
Total	6,613		5.15	2728
Female births				

average=744

MALE POPULATION PROJECTION

Male children born (from Female Life Table)

$$B^M(t) = B(t) \times \frac{SRB}{1+SRB}$$

$$S(0,5) = B^M(0) \times \frac{L(0,5)}{5\ l(0)}$$

Male Life Table

$$S(x,5) = \frac{L(x+5,5)}{L(x,5)}$$

$$N^{t+5}(x+5,5) = N^t(x,5)S(x,5)$$

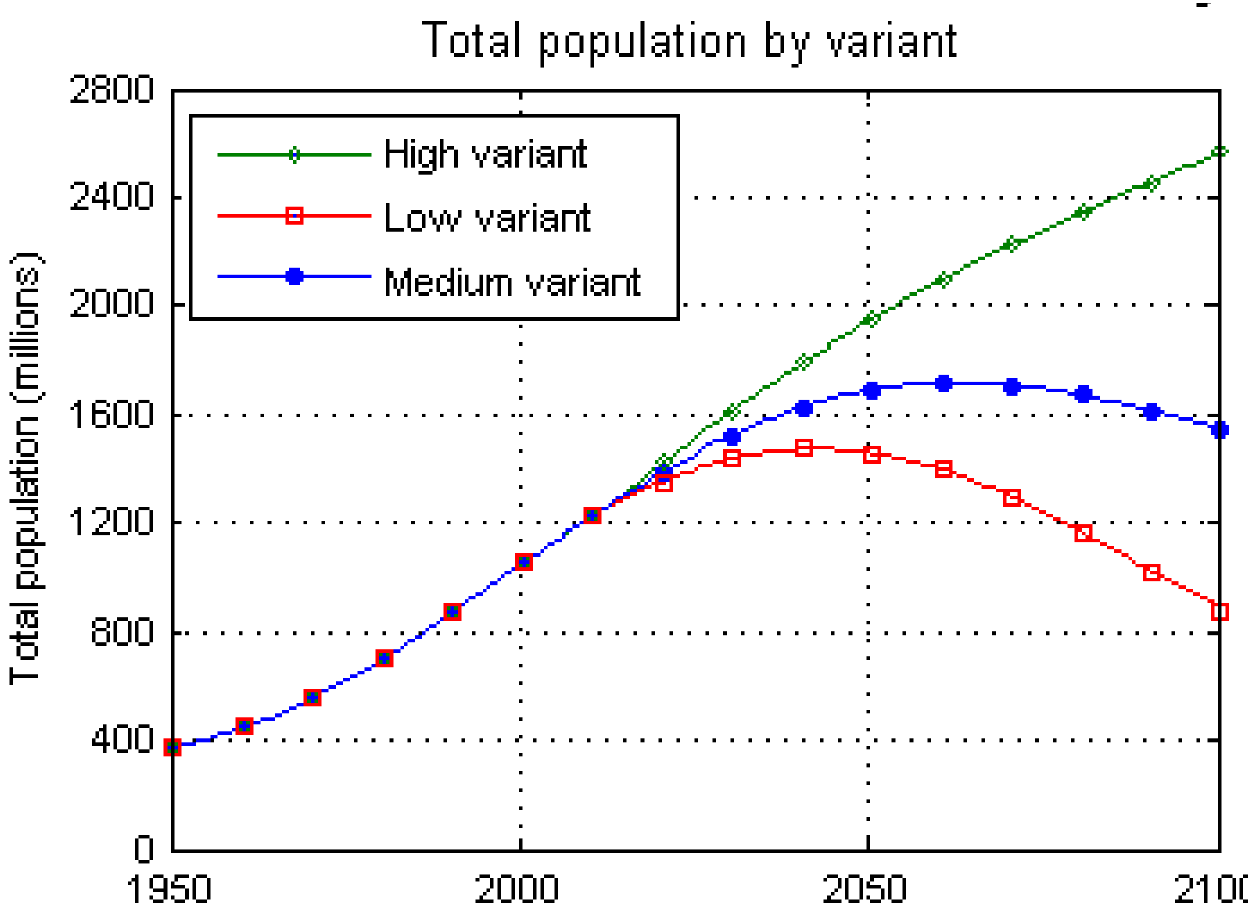
$$0 \le x \le 75$$

Last age : Open Interval

$$N^{t+5}(x+5,\infty) = \left(N^t(x,5) + N^t(x+5,\infty)\right) \times \frac{T(x+5)}{T(x)}$$

$$x = 80$$

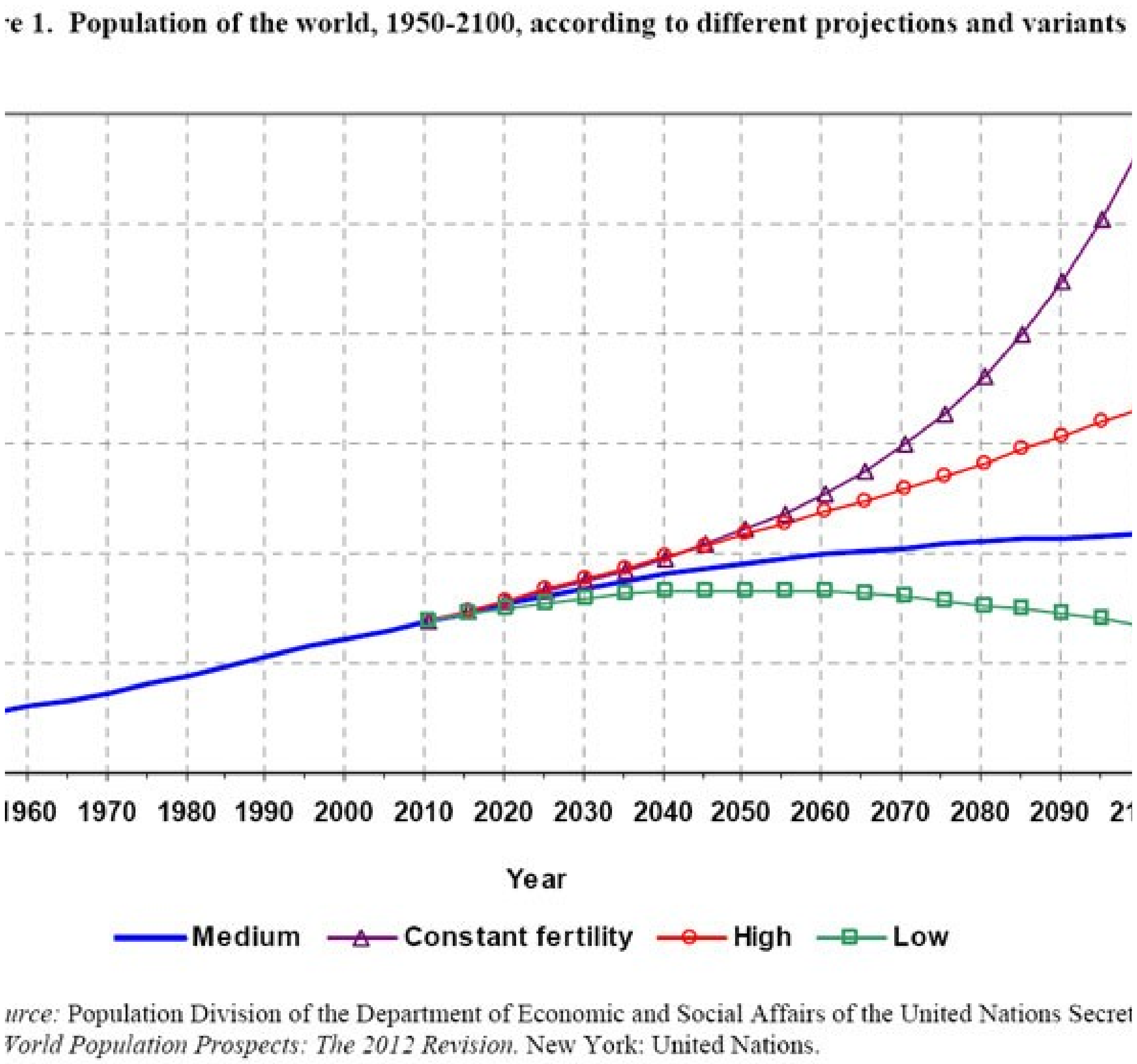
$$T(80) = L(80,5) + L(85,\infty) \quad T(85) = L(85,\infty)$$



UN population Division, Country Profiles- India
http://esa.un.org/wpp/country-profiles/country-profiles_1.htm

WORLD POPULATION PROJECTIONS

- http://www.unfpa.org/pds/trends.htm



ESTIMATES AND PROJECTIONS OF LEB- MEDIUM VARIANT

Table 6.2: *United Nations estimates and projections of mortality levels*

Countries	*Life expectancy at birth (females)*		
	1995–2000	2015–20	2040–50
Nigeria	51.5	57.7	70.2
India	62.9	70.3	76.6
Brazil	71.0	75.6	79.7
China	72.0	76.9	81.0
Germany	80.2	82.2	84.5
USA	80.1	82.1	84.4
Japan	82.9	84.5	86.7

Source: United Nations, 1999. Medium variant.

UN TFR MEDIUM VARIANT

Countries	*Total fertility rate*		
	1995–2000	2015–20	204
Nigeria	5.15	3.52	2.
India	3.13	2.10	2.
Brazil	2.27	2.10	2.
USA	1.99	1.90	1.
China	1.80	1.90	1.
Japan	1.43	1.68	1.
Germany	1.30	1.51	1.

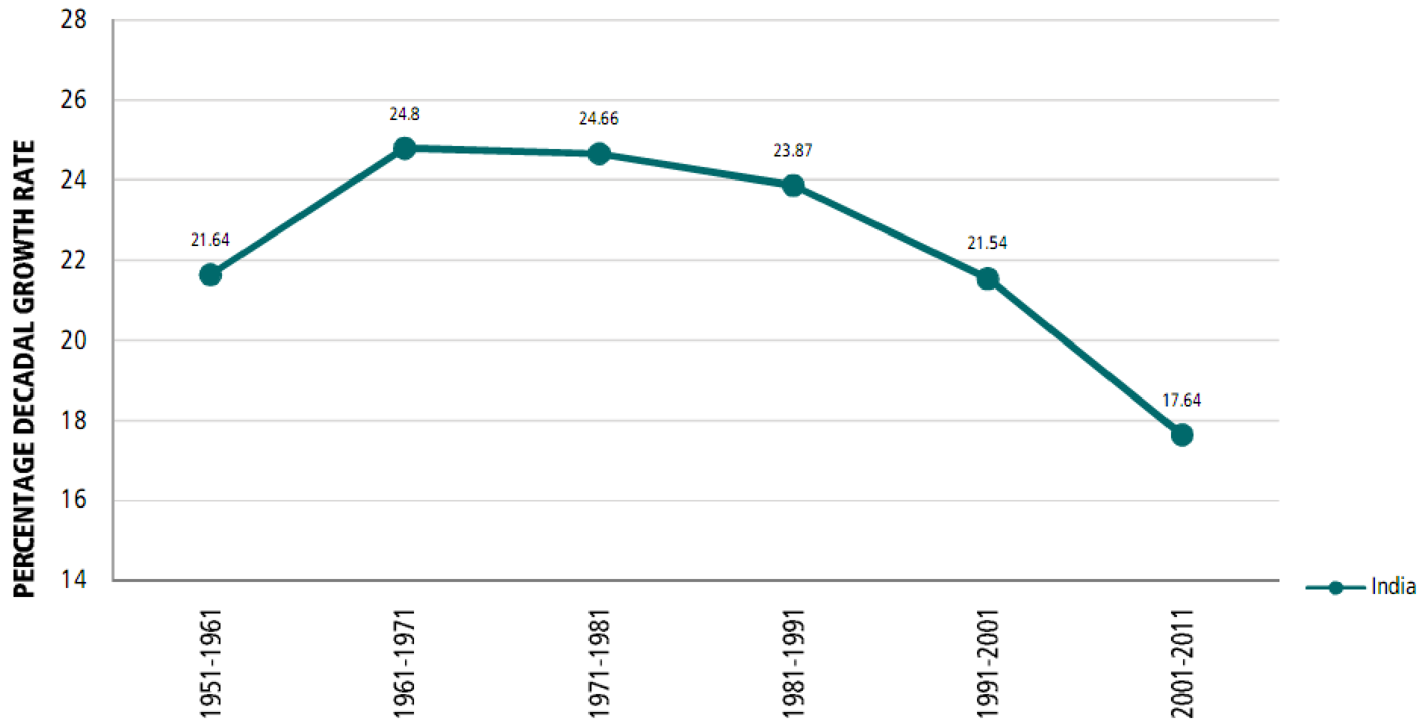
Source: United Nations, 1999. Medium variant.

Table 6.4: *US Bureau of the Census projection assumptions*

Parameter	1995 (*estimate*)	2050 *level (assumption)*		
		Low	Middle	High
TFR	2.055	1.910	2.245	2.580
e_0	75.9	74.8	82.0	89.4
Migration	820	300	820	1370

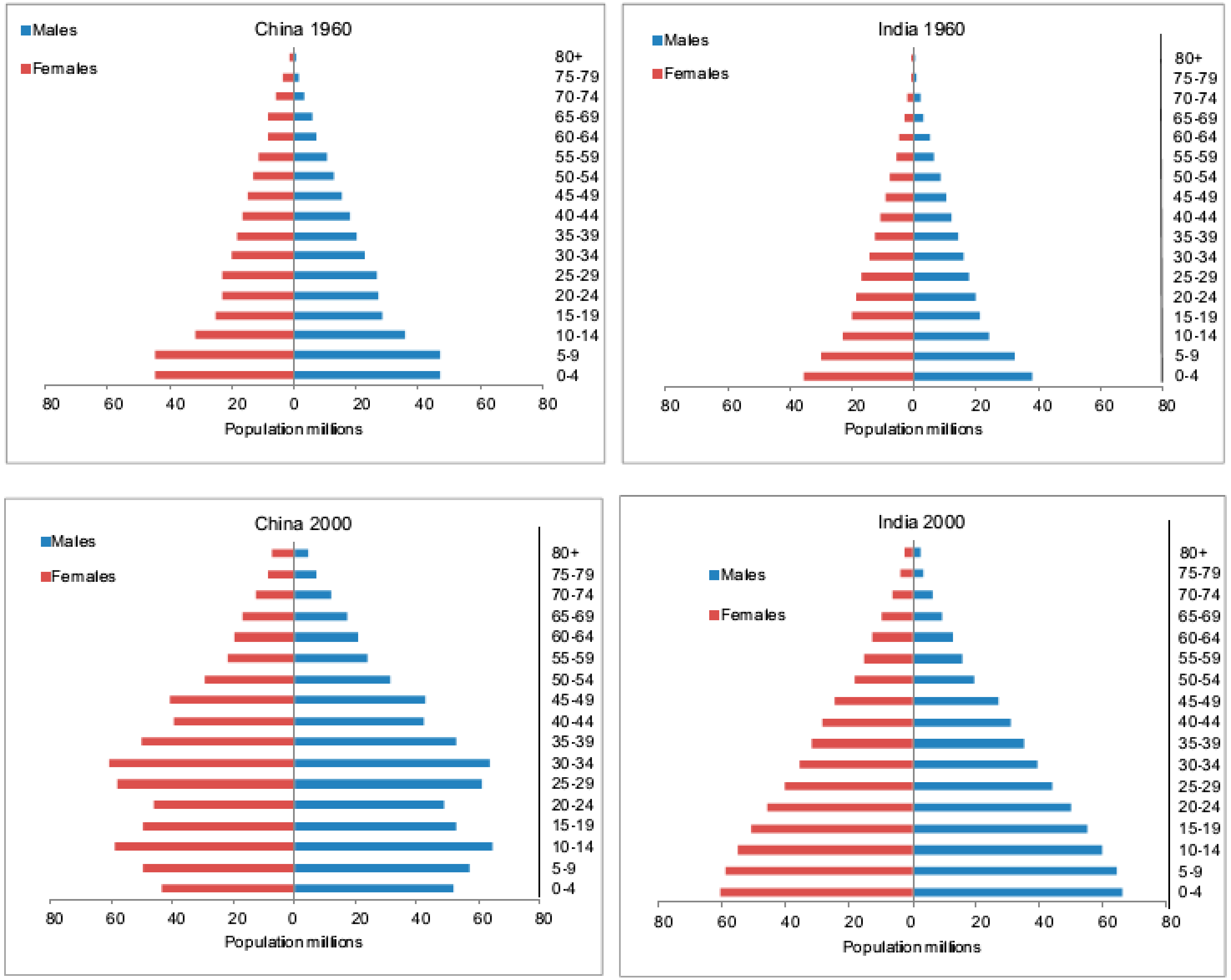
(Migration figures are net immigrants per year in thousands.)

US CENSUS PROJECTION ASSUMPTIONS



Census of India
http://censusindia.gov.in/2011-prov-results/data_files/india/Final_PPT_2011_chapter3.pdf

Figure 2. Comparative Evolution of Population Pyramids



POPULATION MOMENTUM

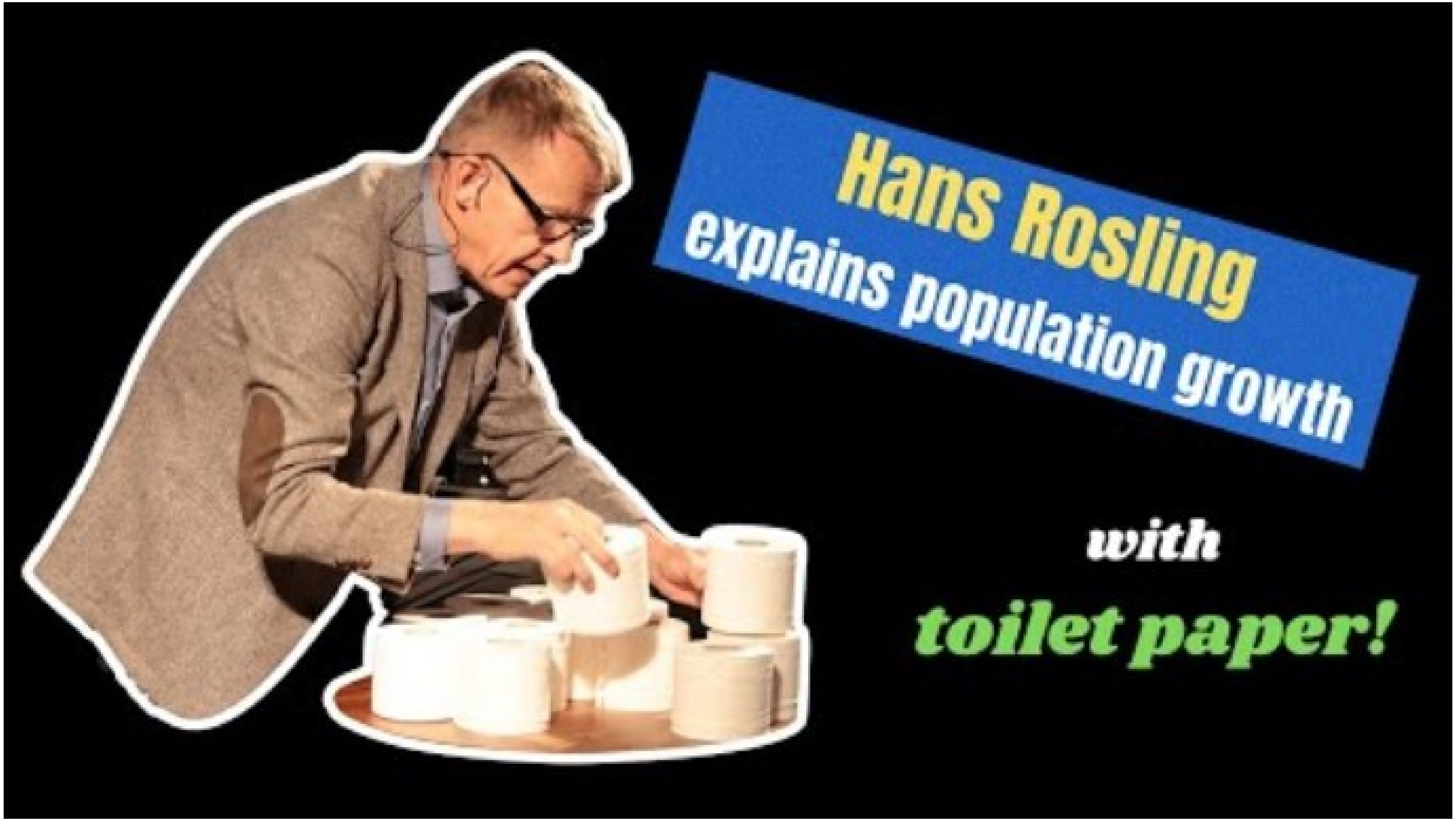
population momentum – the additional growth a population experiences *after* the fertility rate falls to replacement levels.

Even when a country's fertility rate declines to or below replacement level (2.1 children per woman), yet the **population size continues to grow due to the age structure of the population.**

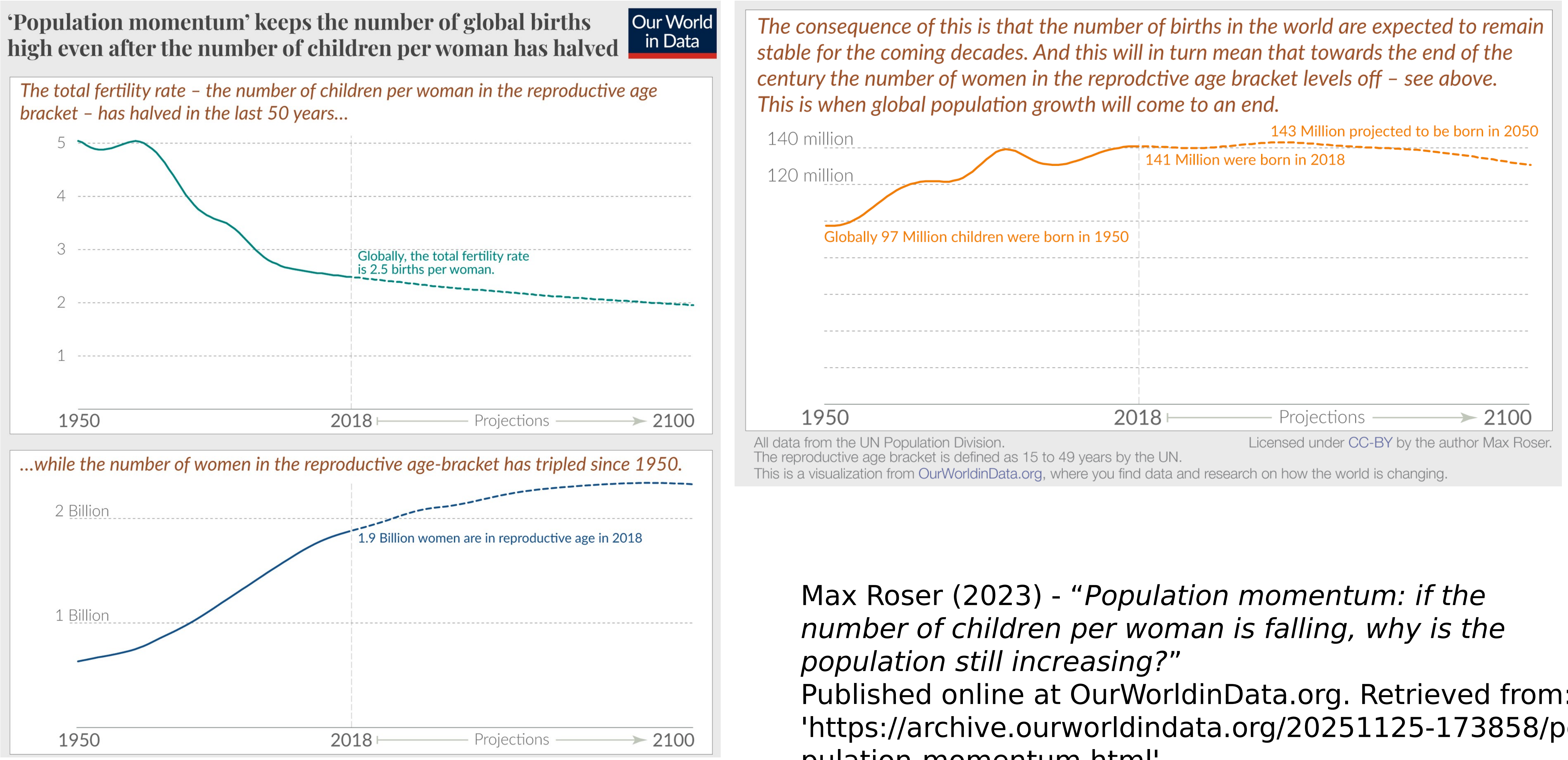
The higher the percentage of young people in a population, the more growth the country will see, even long after fertility rates drop (assuming infant mortality and net migration factors are constant).

As a large number of young people enter their reproductive years, they will have their own children. Even if this generation continues a fertility rate trend below replacement levels, the sheer number of women bearing children will outnumber the overall deaths and keep the population growing for 30-60 years

https://populationeducation.org/population-momentum-why-populations-keep-growing/

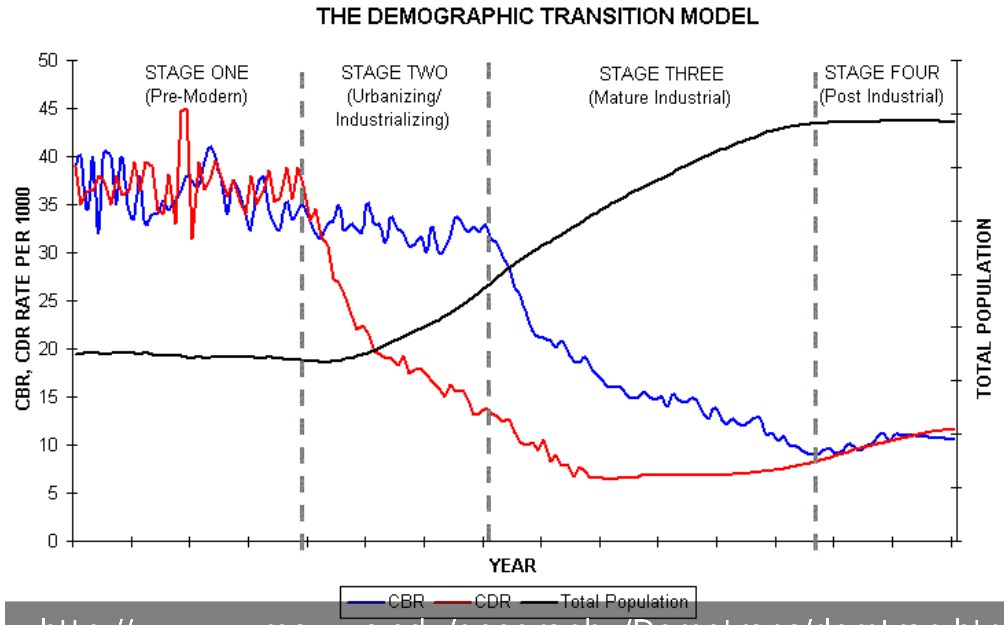


https://www.youtube.com/watch?v=dnmDc-oR9eA

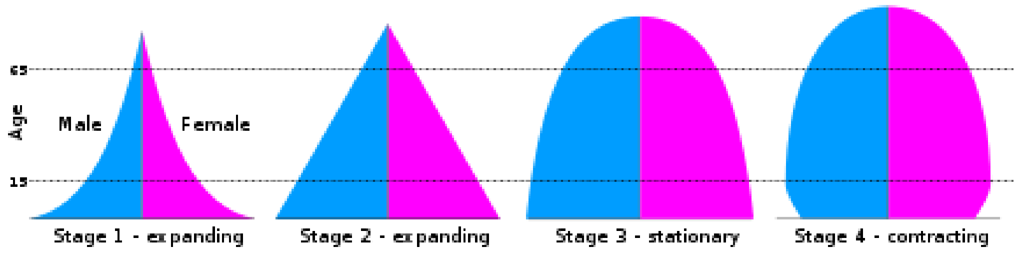


Max Roser (2023) - "*Population momentum: if the number of children per woman is falling, why is the population still increasing?*"
Published online at OurWorldinData.org. Retrieved from: 'https://archive.ourworldindata.org/20251125-173858/population-momentum.html'

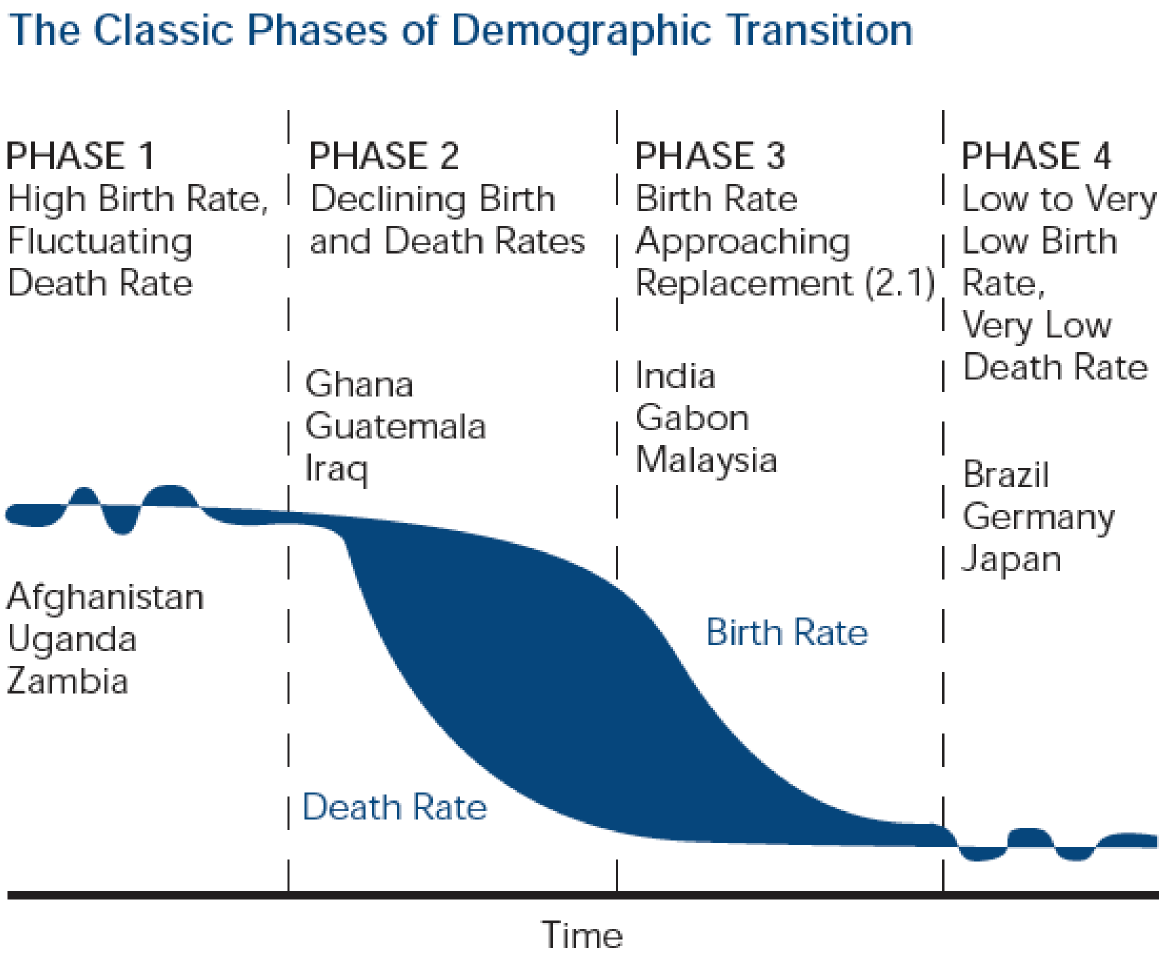
DEMOGRAPHIC TRANSITIONS



http://www.uwmc.uwc.edu/geography/Demotrans/demtran.htm



DEMOGRAPHIC TRANSITIONS

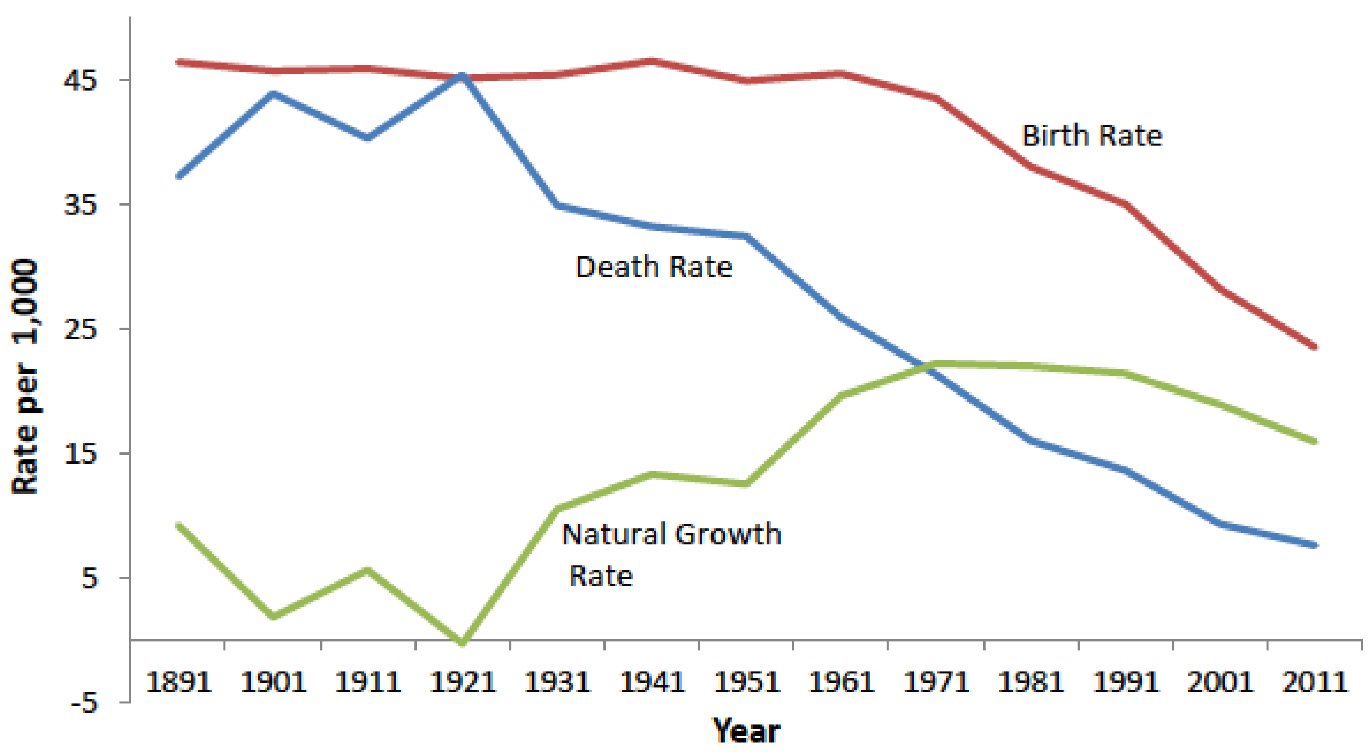


The demographic transition, on the other hand, refers to the transition from high birth and death rates to low birth and death rates

Carl Haub and James Gribble, "*The World at 7 Billion*," Population Bulletin 66, no. 2 (2011)

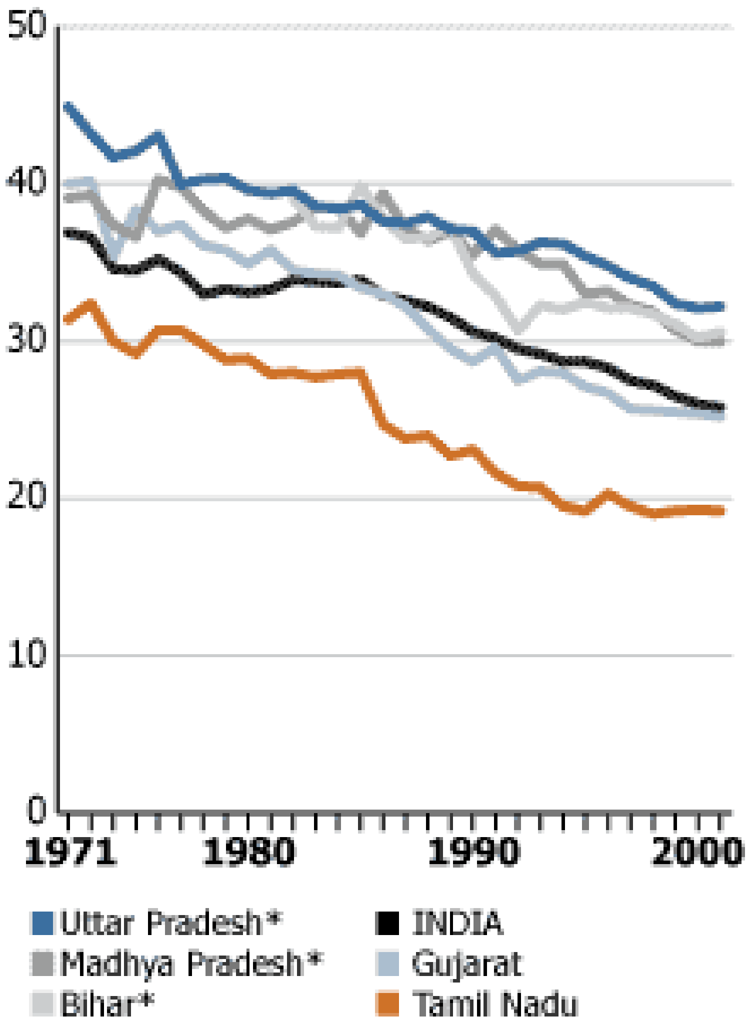
CBR AND CDR - INDIA

Figure 8: The Demographic Transition of India



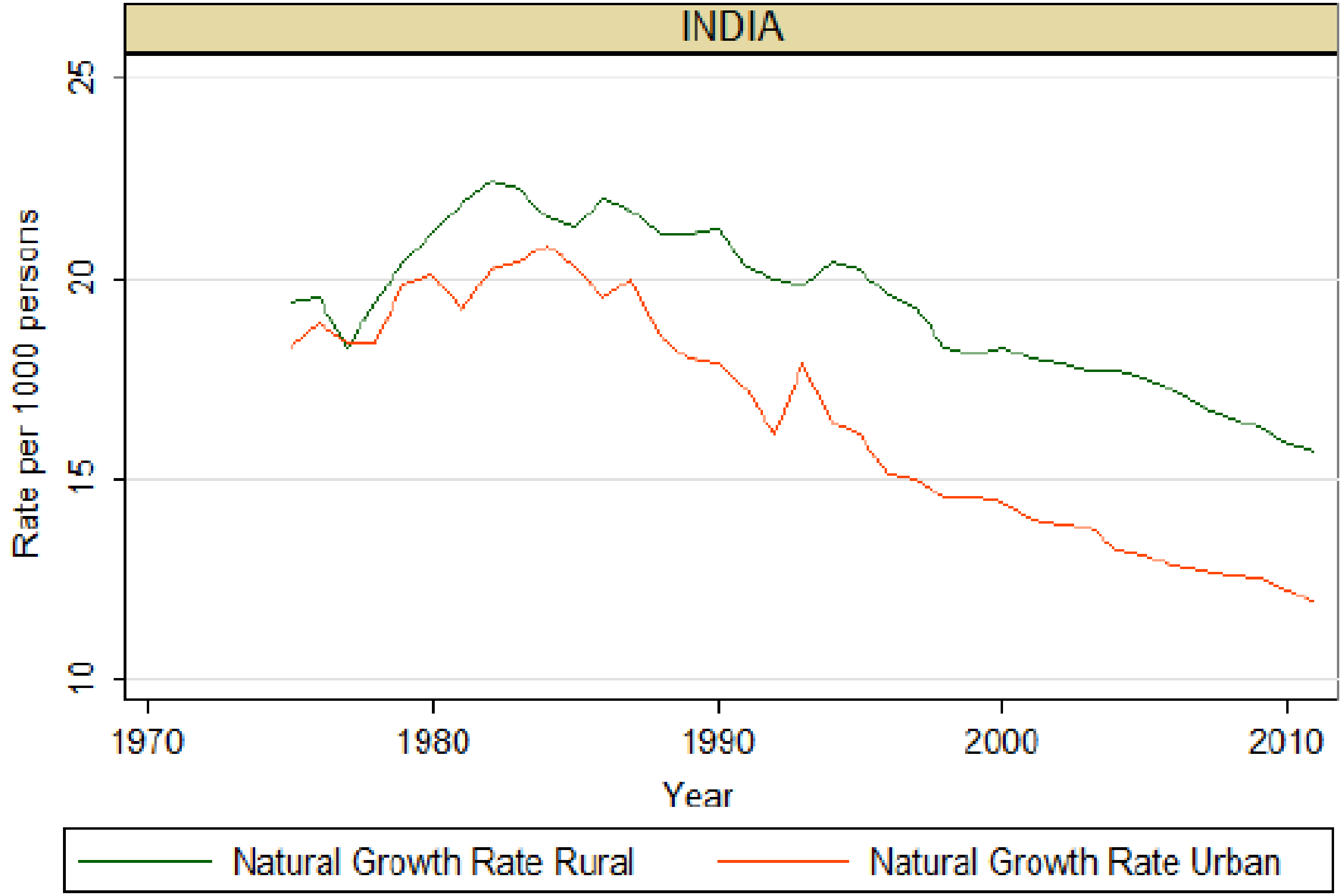
Source: Dyson (2004) and SRS statistics.

Births per 1,000 population



Carl Haub, PRB publications, 2002

Figure 10: Rural-Urban Demographic Divergence of Natural Growth Rates, 1975-2011



Source: Government of India (2009) and bulletins of the Sample Registration System, 2007-2012.

Urbanisation, demographic transition, and the growth of cities in India, 1870-2020

Chinmay Tumbe, 2016

International Growth Centre

NORTH SOUTH DIVIDE

Figure 11: Demographic Divergence in the Southern States, 1975-2011

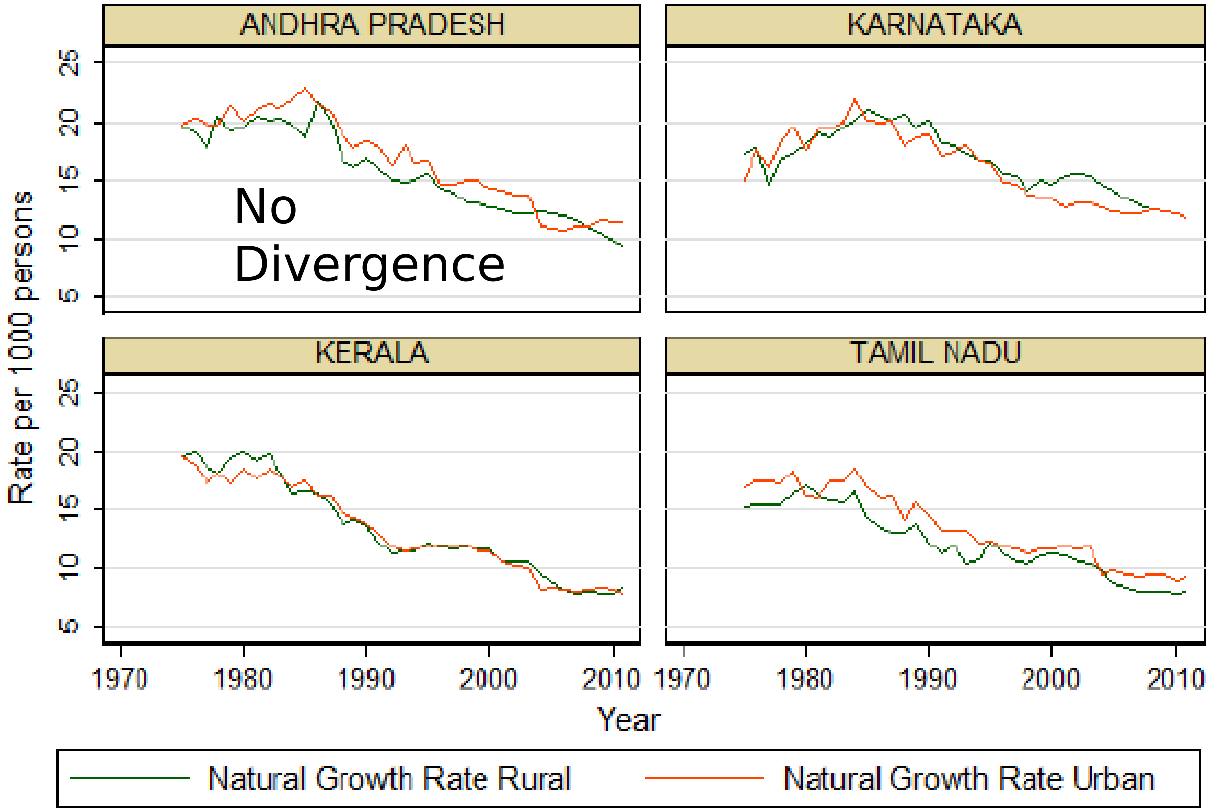


Figure 12: Demographic Divergence in Selected Northern States, 1975-2011

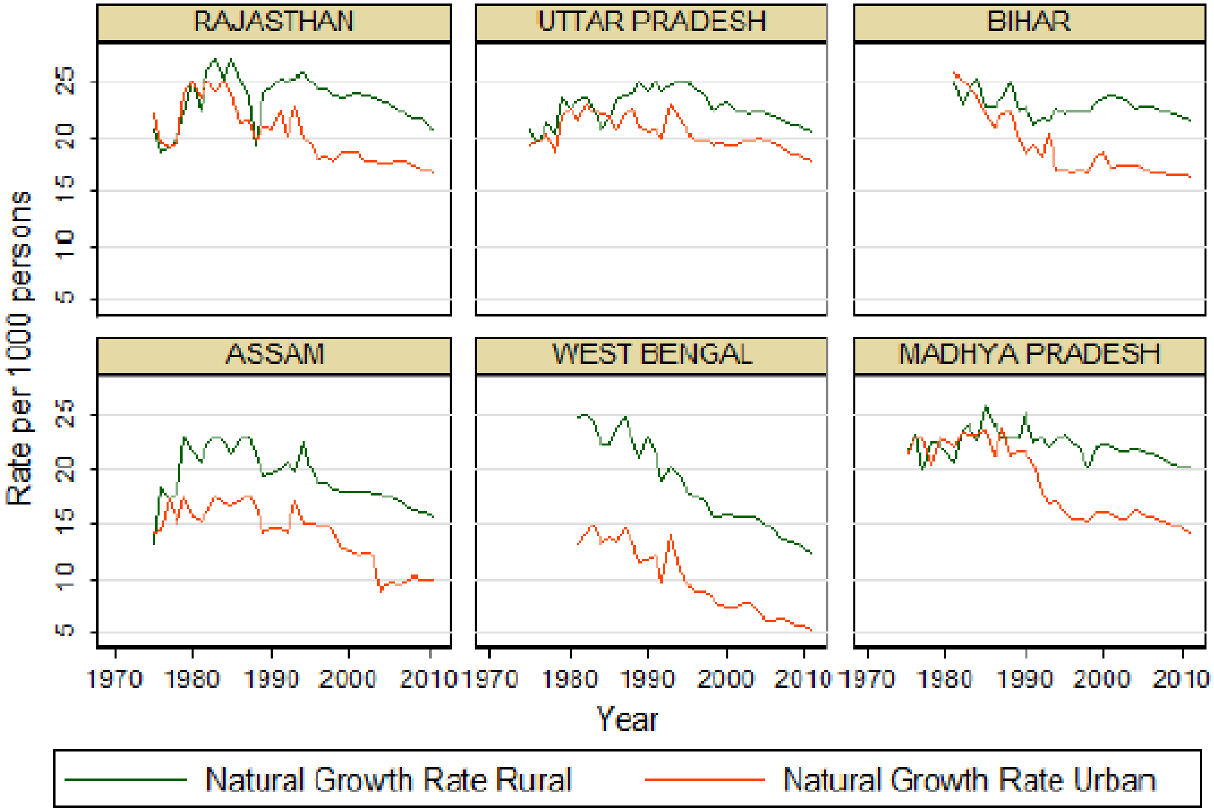
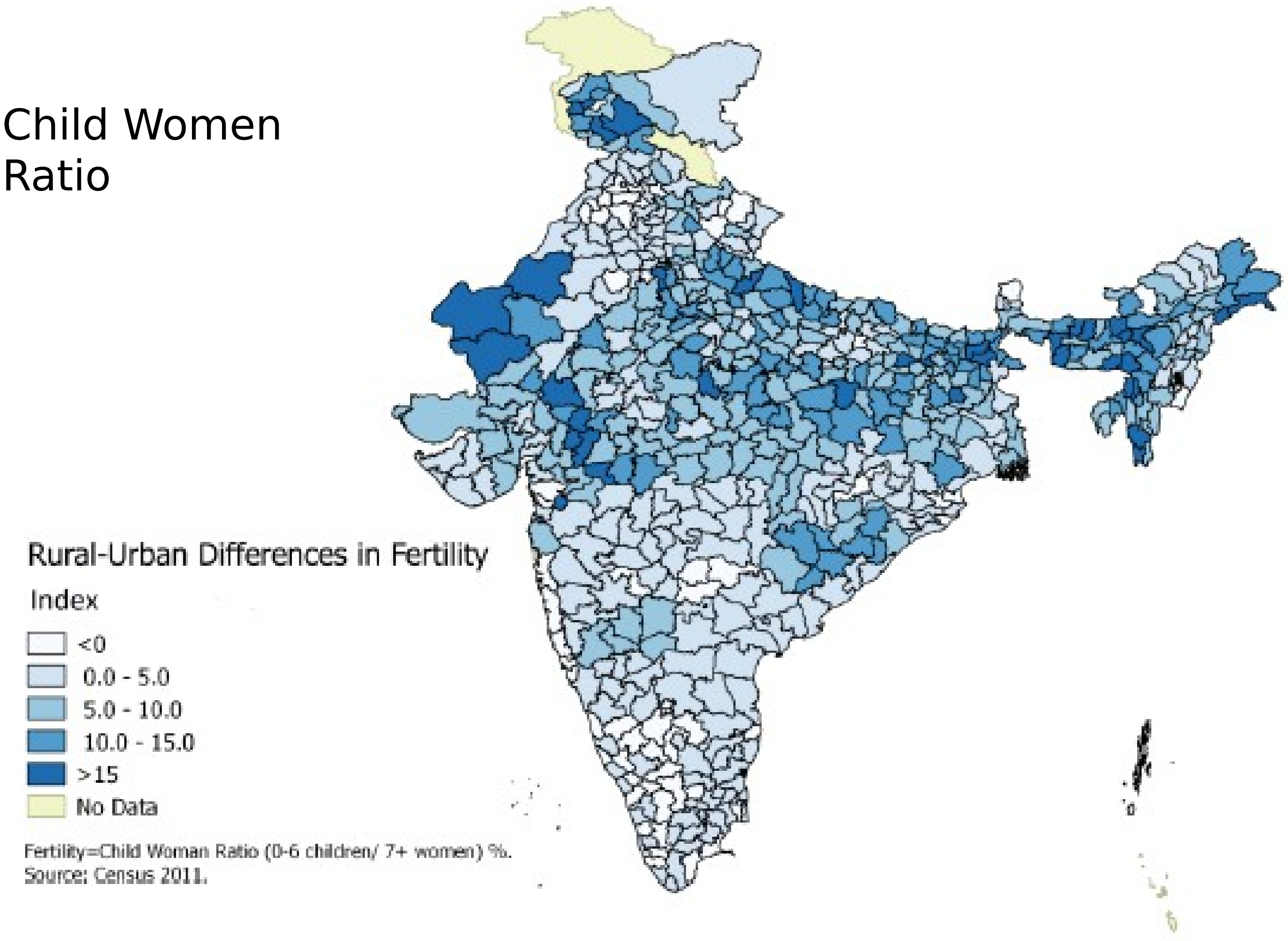


Figure 13: Rural-Urban Differences in Fertility at the District Level

Child Women Ratio



Source: Census 2011.

Table 5: Impact of the Demographic Divergence on Urbanization Rates

Dependent Variable: Urbanization %	(1)	(2)	(3)
Rural-Urban Natural Growth Rate Difference (10 year lag)	-0.68	-0.71	-0.56
	(0.213)***	(0.214)***	(0.202)***
Urbanization % (10 year Lag)		-0.19	-0.12
		(0.164)	(0.152)
Log Number of Towns			6.54
			(1.783)***
Constant	30.07	34.62	5.59
	(1.264)***	(4.208)***	(8.79)
Observations	105	105	105
Time (Decade) Fixed Effects	Yes	Yes	Yes
State Fixed Effects	Yes	Yes	Yes

Notes: Panel data of Indian States at 5-time period intervals: 1971, 1981, 1991, 2001, 2011. Source: Various Census volumes and SRS statistics. Standard errors reported in parentheses. ***Significant at 1% level and **Significant at 5% level.

India's low and slow pace of urbanization thus hinges on three important factors

Lowness: occurs partly because of the nature of India's urban definition and

Slowness: occurs because of

- the *demographic divergence*
- *the highly gendered nature of work- related migration that leads to considerable return migration from cities to villages.*

Figure 14 (A): Population of large UA's in 1901 (in 2011 boundaries)

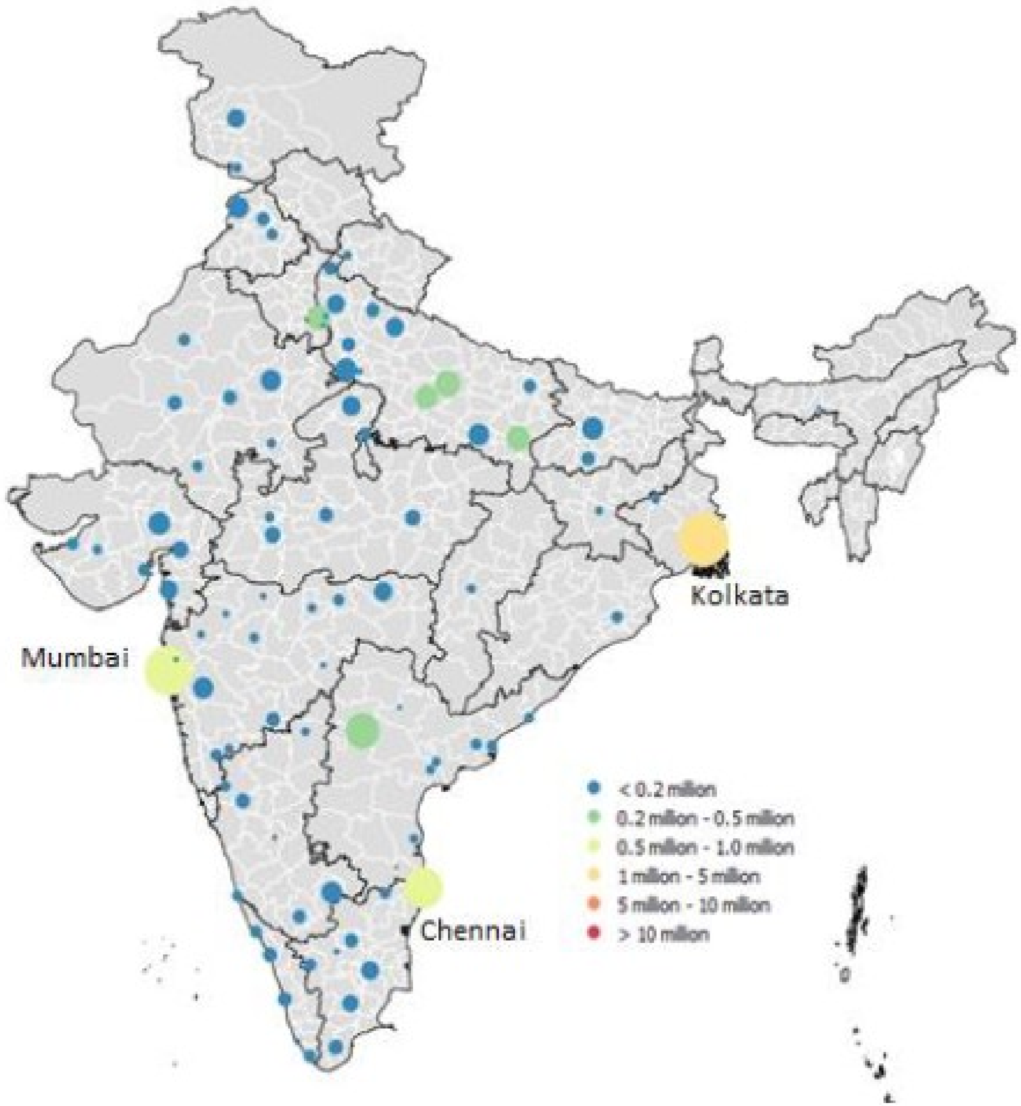
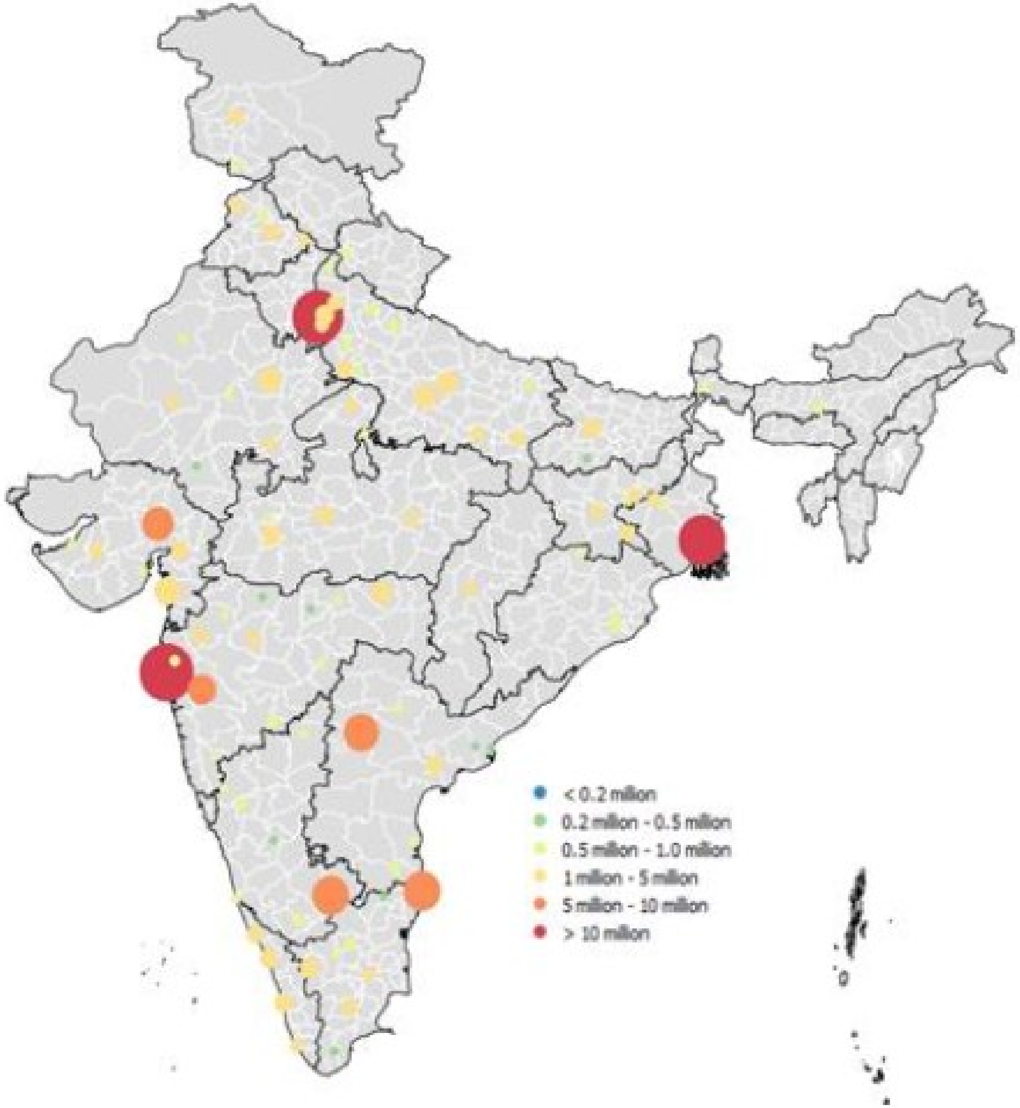


Figure 14 (B): Population of Large UA's in 2011



Source: Various Census Volumes. Top 100 Urban Agglomerations in 2001. City bubbles scaled to population size.

Thus, cities could be classified into four categories –

- **Internal growth** (with minimal migration),
- **External growth** (with minimal fertility)
- **Explosion** (with high fertility and migration)
- **Stagnation** (with low fertility and migration). The

City Growth

Fertility

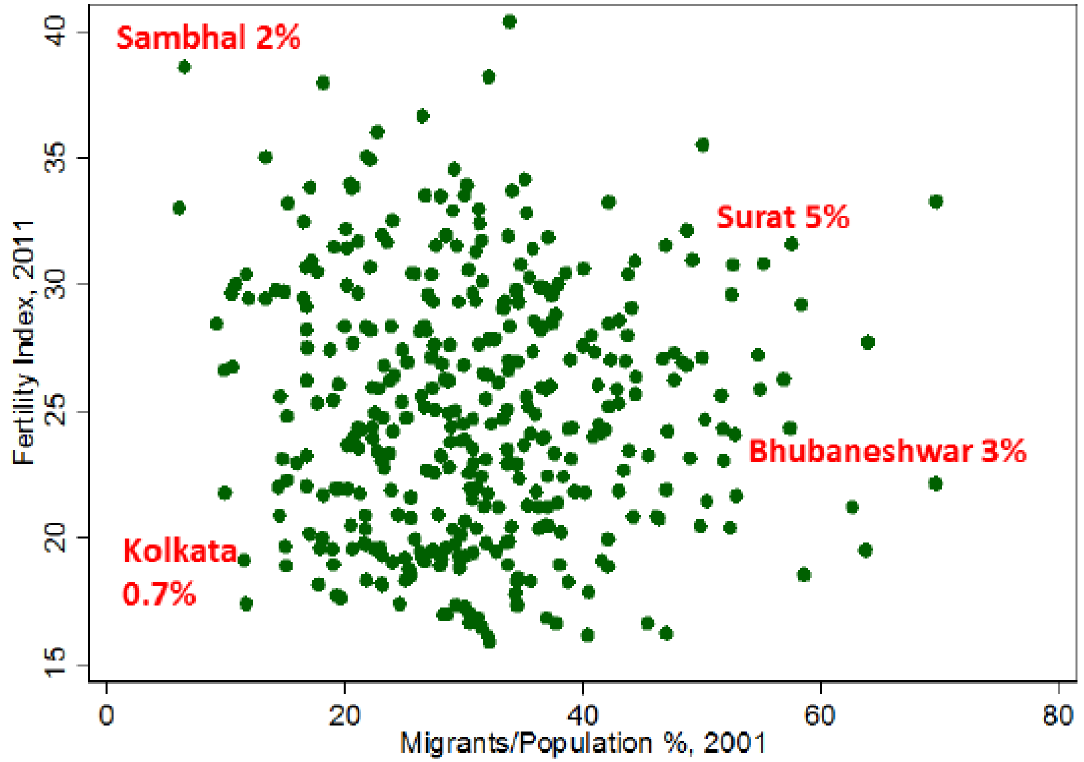
(Human Capital)

High	Internal Growth	Explosion
Low	Stagnation	External Growth
	Low	High

Migration

(Human Capital, Wage Differentials, Gender, Networks)

Figure 18: Fertility vs. Migration across Indian Urban Agglomerations



Source: Census 2011. N= 350+ urban agglomerations. Figures next to marked cities reflect annual growth rate between 2001 and 2011. Fertility Index = % 0-6 Age Children / Women age 7+.